



A synthesis of luminescence and ^{14}C dated dust mass accumulation rates for loess-palaeosol sequences from the Middle Danube Basin

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The Middle Danube (Carpathian) Basin is considered to encompass some of the thickest and most complete records of aeolian dust deposition in Europe, covering the last two glacial–interglacial cycles. In this study, independent, published chronologies based on absolute dating methods were compiled for 34 loess-palaeosol sequences and modelled using Bayesian statistics. Mass accumulation rates (MARs) calculated from the Bayesian age models range from 50 to 1922 $\text{g m}^{-2} \text{ a}^{-1}$ with a mean value of $317 \pm 35 \text{ g m}^{-2} \text{ a}^{-1}$. The glacial–interglacial changes of the MAR estimates only partially agree with the accepted dust deposition model, where high accumulation rates are observed during the cold glacial stages, while low values are typical for the warmer interglacial periods. There is a rapid increase of MAR values during Marine Isotope Stage (MIS) 2, with the highest deposition in most cases observed ~25–18 ka. During MIS 3, the dust deposition was 1.8 times higher than during MIS 4. Regionally, the MARs appear to be highest in the northern part of the Middle Danube Basin, showing decreasing values towards the south. However, due to a variety of factors such as low dating resolution, age distribution, and biases in the dating method, it is debatable whether these results are representative estimates of atmospheric dust activity in the Middle Danube Basin during MIS 5–MIS 1.

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Loess-palaeosol sequences (LPSs) and loess like sediments are considered as some of the most relevant terrestrial archives of climatic change and past atmospheric dust activity. Loess is usually defined as an unstratified, geologically recent aeolian (windborne) deposit of silty or loamy material derived largely from weathered rock surfaces affected by frost shattering or from glaci-fluvial-fluvial deposits of river flood-plains (Pécsi & Richter 1996) and is accumulated mainly during cold glacial phases (e.g. Pécsi 1990). The intercalated palaeosols develop on loess (or into older loess depending on the site) during warmer and generally more humid interglacial periods (Kemp 2001). Since the accumulation process is considered quasi-continuous, LPSs provide important and detailed information on how palaeoclimate change affected the regions where they were accumulated (e.g. Kukla 1977; Porter & An 1995). Loess-like sediments on the other hand, refer to deposits that resemble loess in appearance and composition but may not meet all criteria to be classified as true loess. Like loess, these sediments are primarily made up of silt-sized particles and have a light, porous structure, often with a yellowish or buff colour. However, they differ in several ways that might include: origin of transport (influenced

by other depositional processes, such as fluvial, lacustrine or colluvial processes), grain-size distribution (broader range of particle sizes), mineralogy, structure and layering, and location (they can also be found in locations without direct glacial influence) (e.g. Muhs 2013). Loess-like sediments are thus considered ‘loess analogues’ and can offer valuable information about past environmental conditions, even though they may lack the uniform aeolian characteristics of true loess.

The Middle Danube (Carpathian Basin) encompasses some of the most extensive and complete LPSs in Europe, thus providing detailed archives of environmental changes and atmospheric mineral dust activity for the last 1 Ma (e.g. Buggle *et al.* 2013; Marković *et al.* 2015). The area is situated at the margin between the Arctic, Atlantic, Continental and Mediterranean climate zones, which is why even small changes in the relative influence of associated atmospheric systems can result in significant impacts on the region’s climate (Ducić & Radovanović 2005). During the Quaternary, extensive deposits of wind-blown dust have accumulated as loess over large areas of the region (e.g. Lehmkuhl *et al.* 2021).

Mineral dust plays an important role in the climate system by interacting with radiation, clouds, and

biogeochemical cycles. In addition, natural archives show that dust cycles experienced variability in the past in response to global and local climate change (e.g. Albani *et al.* 2015). Palaeodust archives suggest that the dust cycles during cold periods in the past were increased by aridity (An *et al.* 1991; Liu & Ding 1998) and wind gustiness (e.g. McGee *et al.* 2010).

The understanding of glacial–interglacial dust flux patterns can be greatly improved by information about past mass accumulation rates (MARs) calculated from accurately dated loess profiles. Furthermore, these calculations can provide geological data that can assist in the validation of palaeodust cycle models (Kohfeld & Harrison 2001; Claquin *et al.* 2003; Derbyshire 2003; Frechen *et al.* 2003; Albani & Mahowald 2019), and can improve radiative forcing calculations in relation to ice-age atmospheric conditions.

For this study, we have compiled numerical chronologies for 34 LPSs from the Middle Danube Basin to investigate the variability (spatial and temporal) and magnitude of dust mass accumulation rates over the last five marine isotope stages for this region. Additionally, we investigated the reliability of the published chronologies, possible biases in the dating methodology and their applicability for MAR determination.

Geological context of the study area

The Carpathian Basin developed as a consequence of back-arc stretching during the formation of the Carpathian Arc (Balázs *et al.* 2016). Several back arc sub-basins were formed in this process: the Vienna Basin, Transcarpathian Depression, West Danube Basin and Transylvanian Basin, which all surround the central, Carpathian Basin (Sclater *et al.* 1980). These basins are the result of tectonic movements during the Alpine Orogeny, starting from the Palaeogene. By the time the Paratethys Sea got separated (11.5 Ma ago, Juhász *et al.* 2007), a lake started to form and its water became gradually brackish (Haas 2015). Due to intense basin subsidence, basaltic volcanism started in some parts of the Carpathian Basin 8 Ma ago and lasted until the beginning of the Quaternary (Nemeth & Martin 1999). Analyses of core-sample data in the Carpathian Basin made a reconstruction of the fluvio-deltaic systems possible (Juhász *et al.* 2007). The fill of the basins originated from the surrounding mountains: the Alps and Dinarids (Schmid *et al.* 2008), including the loess material, which was probably transported by rivers and subsequently accumulated by winds (Bugge *et al.* 2008; Újvári *et al.* 2008; Újvári *et al.* 2012). Several studies suggest that the source material for the formation of the loess cover in the Danube Basin came from short-distance aeolian transport, and that the processes of sediment mixing occurred in the alluvial plains of the main rivers, and not by aeolian transport in the air (Bugge *et al.* 2008; Smalley *et al.* 2009; Fenn *et al.* 2022).

Material and methods

Site selection

The LPSs for this study were chosen from within the boundaries of the former permafrost region (both continuous and discontinuous permafrost) in the Middle Danube Basin. The permafrost boundaries were constructed according to van Vliet-Lanoë *et al.* (2004) and are presented in Fig. 1. The chronologies were attained from published literature and categorized according to their geomorphological setting following the classification of Kohfeld and Harrison (2001) and Újvári *et al.* (2010). Four main geomorphological settings were distinguished: loess plateau, river terrace/river bank, hill slope/crest and plain. In this study, loess plateaus are represented by 15 sites (e.g. Madaras, Batajnica, Vukovar), plain settings by three sites (Debrecen, Lakitelek brickyard and Szeged-Óthalom), hill slopes by eight sites (e.g. Bodrogkeresztúr I, Tokaj Kereszt Hill, Irig) and river terraces/banks by eight sites (e.g. Dunaszekcső, Belotina, Požarevac) (Table 1). Here it is important to highlight the distinction between loess plateaus and loess plain sites, as the difference is sometimes not so obvious. Loess plateaus present elevated landforms with thick, continuous loess deposits, typically accumulating from long-term aeolian processes. These sites often preserve stable sedimentary records with minimal reworking, making them valuable for reconstructing past environmental conditions. Loess plain sites on the other hand are usually flat, low-lying areas with loess deposits, often thinner and potentially more influenced by reworking or interruptions due to proximity to rivers and flood-plains. Such sites can reflect mixed depositional influences, adding complexity to palaeoenvironmental interpretations (Pécsi 1990). These distinctions highlight differences in depositional continuity, sediment thickness, and exposure to reworking, all of which are relevant to understanding loess records in varying topographical contexts.

The local geomorphological setting was determined in regards to the digital elevation model and the slope map resulting from it. Data used to obtain the values of the digital elevation model derive from the interpolation of digitized contour lines and photogrammetry. The vertical accuracy of the interpolated values is 5.0 ± 0.1 m, covering Hungary, Serbia and Croatia (Open Maps for Europe Europeographics n.d.). Slope analysis refers to the modern topographic surface. LPSs with an inclination $>10^\circ$ were categorized as slope sites. The highest inclination values were obtained at Tokaj hill in Hungary, where the slopes were calculated via ArcGIS Pro and were determined to be up to 17° . Here we note that these data refer to the present-day settings. Currently, it is not possible to determine the palaeo-slopes at the time of the loess accumulation; however, we can assume that the inclinations had similar values. The slope data are presented in Table 1. Although the site distributions are

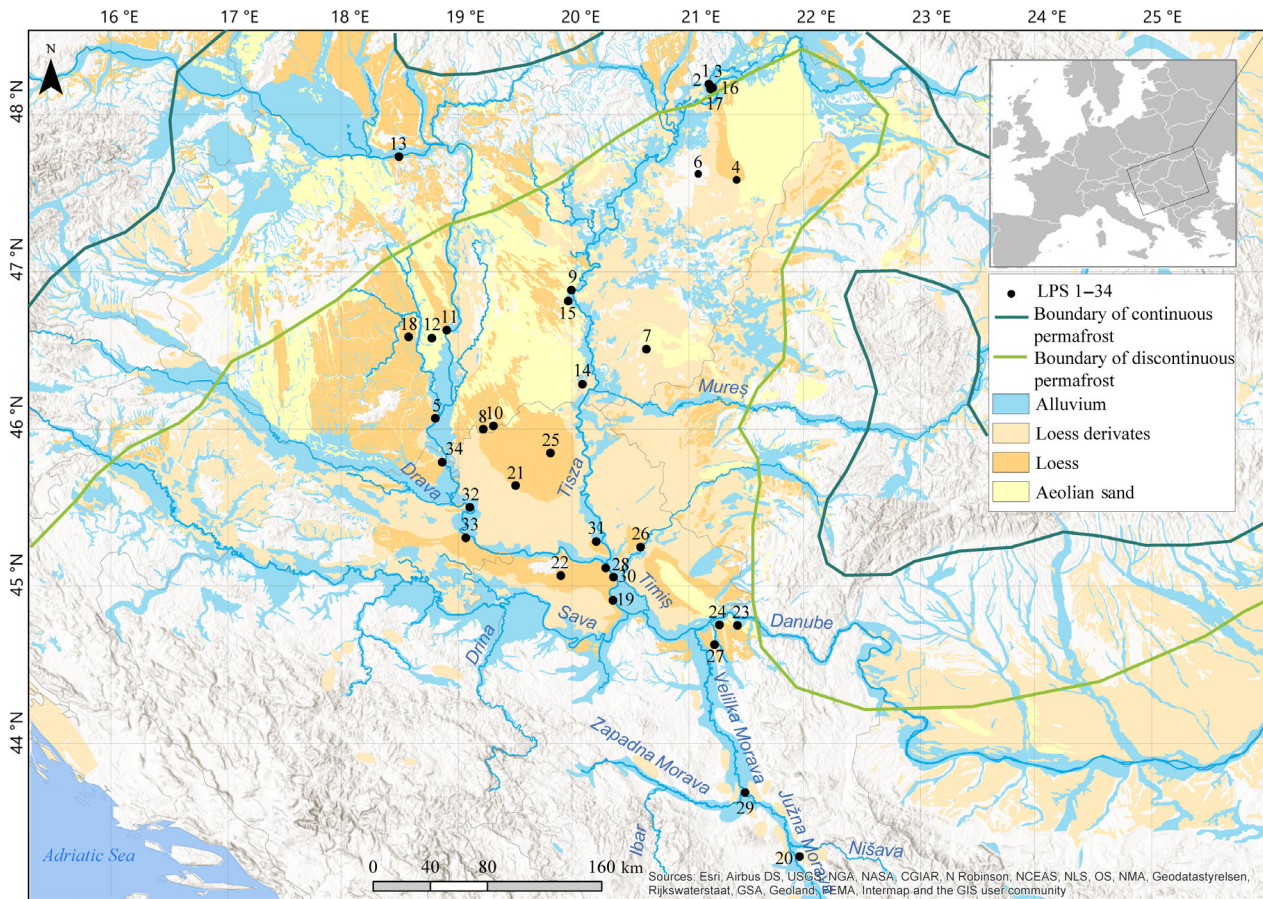


Fig. 1. Map of the Middle Danube Basin showing locations of the investigated LPSs (Lehmkuhl *et al.* 2021, modified). The permafrost boundaries are constructed according to van Vliet-Lanoë *et al.* (2004). See Table 1 for more information on the loess profiles included.

not related to national borders, for a better overview, we will also be referring to the countries where the individual sites are located (Table 1). Almost all investigated LPSs are positioned in the vicinity of the Danube and Tisza rivers. The exceptions are two southernmost profiles in Serbia, Belotinac and Stalać, which do not have links to the mentioned rivers, and are associated with the Velika Morava and Južna Morava rivers. The geographical positions of the individual LPSs are presented in Fig. 1 and Table 1. In most studies, the LPSs are classified according to their geomorphological settings (Table 1) as specified in the DIRTMAP 3.0 database documentation (Kohfeld & Harrison 2001; Újvári *et al.* 2010). This is because distinct aeolian sedimentation is expected under different geomorphological situations. However, one factor that is rarely addressed in loess mass accumulation rate studies is the proximity of loess profiles to potential dust sources (ultimate and immediate). It is known that the restructuring of large river networks, as well as changes in sediment load and discharge, can considerably affect the dust accumulation rates at nearby loess depo-centres (Marković *et al.* 2015; Perić *et al.* 2022a). Hence, in this study, we include information on the

distance of the individual LPSs from the Danube and Tisza rivers (immediate dust sources; Table 1). Conversely, additionally to these main dust sources, there are also areas covered with loess derivatives, aeolian sands and alluvium. Loess derivatives refer to materials that are formed from the alteration, reworking, or modification of loess deposits, often due to processes like soil formation, erosion, or secondary transport by water. These derivatives may have different textures or mineral compositions compared to the original loess (Pécsi 1990). Aeolian sand areas are regions where sand has been transported and deposited by wind activity. These areas are often characterized by well-sorted, rounded sand grains due to consistent wind action, and they can form distinct landforms such as dunes, sand sheets, and sand ridges. Aeolian sands play an essential role in understanding past climates, as their distribution and characteristics can indicate wind patterns and aridification trends in Earth's history (Pye & Tsoar 2009). Alluvium refers to loose, unconsolidated sediments (such as clay, silt, sand and gravel) that have been deposited by running water, typically in riverbeds, flood-plains, and deltas. These deposits accumulate over time through the action of rivers

Table 1. Information on the investigated sites including: geographical position (longitude and latitude), elevation above main sea level (a.m.s.l.), geomorphological setting (landform), slope angle and distance to the immediate, and potential secondary dust sources. If distance value is 0.00, LPS is located within indicated category. LPS site numbers correspond to the geographical locations shown in Fig. 1.

Site	Country	Longitude	Latitude	Elevation a.m.s.l. (m)	Landform	Slope (°)	Distance to Danube (km)	Distance to Tisza (km)	Distance to loess (km)	Distance to plateaus (km)	Distance to loess derivates (km)	Distance to aeolian sand (km)	Distance to alluvium (km)
1. Bodrogresztúr I	Hungary	48°07'37"N	21°33'58"E	109	Hill slope	14.0	172.7	4.3	0.0	0.0	4.8	7.4	0.3
2. Csorgókút I	Hungary	48°07'31"N	21°24'07"E	150	Hill slope	13.4	174.4	0.9	0.2	0.2	6.4	5.1	0.3
3. Csorgókút II	Hungary	48°07'32"N	21°24'12"E	132	Hill slope	14.6	172.9	2.2	0.3	0.3	5.0	4.3	1.2
4. Debreceen brickyard	Hungary	47°31'50"N	21°34'13"E	n/a	Plain	0.3	184.7	48.6	0.8	0.8	0.0	1.3	13.5
5. Dunaszekeső	Hungary	47°31'50"N	21°34'13"E	106	River bank	6.4	0.5	98.6	0.5	0.5	7.1	10.8	0.0
6. Hortobágy, Szálkahalom	Hungary	47°35'01"N	21°09'07"E	~87	River terrace	0.3	159.3	26.4	13.0	3.3	3.3	19.3	18.5
7. Kardoskút, Fehér- lake	Hungary	46°29'42"N	20°41'42"E	~88	River terrace	0.7	131.3	37.6	28.1	0.7	0.7	0.0	12.3
8. Katymar	Hungary	46°01'03"N	19°11'43"E	107	Loess plateau	5.3	27.8	64.9	0.0	0.0	3.8	0.2	0.7
9. Lakitelek brickyard	Hungary	46°52'55"N	20°01'27"E	88	Alluvial fan	0.3	80.1	2.3	0.0	0.0	2.3	0.3	0.4
10. Madaras	Hungary	46°02'14"N	19°17'15"E	132	Loess plateau	4.7	32.5	57.8	0.1	0.1	10.9	0.9	0.0
11. Paks I core	Hungary	46°38'46"N	18°52'30"E	147	River terrace, loess plateau	5.7	0.8	91.4	0.0	0.0	1.8	2.7	0.5
12. Paks II core	Hungary	46°35'40"N	18°44'17"E	147	River terrace, loess plateau	1.6	9.6	103.0	7.4	7.4	10.3	0.0	1.6
13. Süttő	Hungary	47°44'24"N	18°26'48"E	256	River terrace	2.0	1.9	145.6	0.0	0.0	46.0	7.3	0.0
14. Szeged-Öthalom	Hungary	47°16'58"N	20°05'48"E	85	Sandy loess dune field	0.9	91.1	5.7	1.6	1.6	0.0	5.6	5.6
15. Tiszaalpári high- bluff	Hungary	46°48'47"N	19°59'38"E	79	River terrace	0.4	76.4	3.9	0.6	0.6	4.4	0.0	0.4
16. Tokaj Kereszt Hill	Hungary	48°07'33"N	21°24'05"E	122	Hill slope	16.8	174.4	0.9	0.2	0.2	6.4	5.2	0.4
17. Tokaj Palkó-quarry	Hungary	48°07'29"N	21°24'14"E	130	Hill slope	14.2	174.5	0.7	0.0	0.0	6.6	5.1	0.1
18. Udvari borehole	Hungary	46°25'25"N	18°47'25"E	~178	Hill slope	10.9	172.7	118.5	0.0	0.0	26.5	4.2	1.5
19. Batajnica	Serbia	45°55'29"N	20°19'11"E	113	Loess plateau	1.8	0.8	24.2	0.5	0.5	3.5	36.7	0.0
20. Belotinac	Serbia	43°15'34"N	21°52'00"E	228	Loess covering river terrace	3.0	103.2	244.2	142.5	142.5	146.3	154.5	0.8

(continued)

Table 1. (continued)

Site	Country	Longitude	Latitude	Elevation a.m.s.l. (m)	Landform	Slope (°)	Distance to Danube (km)	Distance to Tisza (km)	Distance to loess plateaus (km)	Distance to loess derivates (km)	Distance to aeolian sand (km)	Distance to alluvium (km)
21. Crvenka	Serbia	43°39'42"N	19°28'43"E	105	Loess plateau	2.2	33.3	45.4	0.0	0.5	40.2	21.2
22. Irig	Serbia	45°05'04"N	19°51'58"E	185	Hill slope	10.8	13.5	32.9	0.0	8.4	66.6	10.1
23. Kisihevo	Serbia	44°44'00"N	21°25'00"E	69	Loess plateau	0.4	5.1	100.4	1.5	4.1	1.3	0.0
24. Nosak	Serbia	44°44'31"N	21°15'28"E	127	Loess plateau	1.5	4.7	88.9	0.0	1.5	3.4	0.3
25. Novo Orahovo	Serbia	45°51'55"N	19°48'01"E	99	Loess plateau	0.3	66.1	22.2	0.0	16.5	25.2	14.3
26. Orlovat	Serbia	45°15'00"N	20°35'00"E	88	Loess plateau	0.5	26.9	21.7	0.2	0.0	13.2	0.0
27. Požarevac	Serbia	44°37'06"N	21°12'25"E	144	Loess slope on river terrace	2.7	14.5	93.3	0.0	6.8	16.9	1.8
28. Stari Slankamen	Serbia	45°07'34"N	20°15'57"E	140	Loess plateau	0.5	1.3	1.5	0.0	0.3	34.8	0.0
29. Stalać	Serbia	43°41'00"N	21°25'00"E	~140	Loess slope on river terrace	6.4	105.5	185.8	92.4	88.0	107.3	0.6
30. Surduk	Serbia	45°04'00"N	20°20'00"E	111	Loess plateau	0.7	1.0	8.9	0.0	1.5	31.2	0.4
31. Veliki Surduk	Serbia	45°18'00"N	20°12'00"E	116	Loess plateau	1.9	12.0	3.7	0.0	0.7	43.0	0.4
32. Erdut	Croatia	45°31'36"N	19°03'53"E	~130	Hill slope	10.5	0.8	77.9	1.0	0.7	45.8	0.0
33. Vukovar	Croatia	45°21'00"N	19°00'12"E	108	Loess plateau	2.5	1.2	86.2	0.0	5.9	60.2	0.0
34. Zmajevac	Croatia	45°48'44"N	18°49'13"E	95	Loess plateau	8.7	1.9	96.8	0.0	1.3	15.5	0.0

and streams and are often rich in organic material, making them fertile for agricultural use (e.g. Bridge 2009). Such areas can often contribute to the dust accumulation of a certain LPS during periods of higher wind activity. To ensure consistent terminology and avoid misunderstanding, we will refer to these areas as ‘potential secondary immediate sources’. Hence, in this study, we also include the distance of the investigated LPSs to these dust sources.

Even though data were available for a variety of LPSs (e.g. Marković *et al.* 2008; Újvári *et al.* 2010), for most of them, the chronology was established using relative dating methods. Racemization of amino acids in shells of certain mollusc genera (*Pupilla*, *Trichia*, *Helicopsis*) preserved in deposited loess served as a relative dating method in some of the oldest LPSs in the Carpathian Basin: Ruma, Požarevac, and Veliki Surduk and Stari Slankamen (Marković *et al.* 2006, 2021; Radaković *et al.* 2023). Together with the magnetic susceptibility of the alternating loess and palaeosols, correlation to the glacial and interglacial cycles, respectively, and the marine isotope stratigraphy (Lisiecki & Raymo 2005) became possible. Although sound in general, this approach can also be very subjective and can potentially lead to grave misinterpretations of the pedostratigraphical record. For instance, the first stratigraphy of the Požarevac LPS was established using field observations, amino acid racemization and magnetic susceptibility-based correlation with the LPS in Vojvodina (Jovanovic *et al.* 2006). According to the proposed chronology, two glacial loess units (L1 and L2) and one interglacial soil were identified. However, a later study by Perić *et al.* (2021), where feldspar pIRIR₂₉₀ dating was conducted, determined that the Požarevac LPS consists of three glacial units (L1, L2 and L3) and two interglacial soils, which suggests that a whole glacial cycle was not recognized in the original paper. Therefore, the first stratigraphy chronologies are considered to have a low reliability and would potentially produce high uncertainties in the dust flux calculations. Hence, in this study, we include only those loess sequences for which absolute chronologies (e.g. ¹⁴C, OSL, IR, pIRIR) are available. Out of the 34 investigated sites, 15 (44%) were dated using ¹⁴C, for seven sites (21%) the high-temperature pIRIR feldspar dating method was employed, four sites (12%) used low temperature IRSL feldspar dating, another four sites used quartz OSL, while for four sites (12%) a combined OSL/pIRIR or OSL/IRSL chronology was constructed. For the MAR investigation, from the available ¹⁴C chronologies, we used only BP (before present) calibrated ages. The IRSL and pIRIR dates employed in this investigation were all corrected for anomalous fading and the residual dose was subtracted (for details see original studies e.g. Böskén *et al.* 2019; Avram *et al.* 2020; Novothny *et al.* 2020; Perić *et al.* 2022b). To ensure a reasonable degree of reliability, dates displaying an age reversal (outside uncertainty) were rejected and not used in the age-depth modelling.

Likewise, if for the same depth, more than one date was available, the age displaying a lower uncertainty was chosen. Based on these criteria, 95 ¹⁴C ages from the Dunaszekcső LPS were excluded from this study (for details see Data S1 and Újvári *et al.* 2017). Unless otherwise stated in the original paper, we consider the published ages to be reliable and suitable for age-depth modelling. Detailed information on the chronologies used is presented in Table 2 and Data S1.

Age-depth modelling and MAR calculation

The available absolute chronologies enabled us to construct independent and continuous age-depth models for each of the investigated LPSs, which consequently result in a more robust calculation of sedimentation rates. There are several age-depth modelling methods that have been adopted and used for constructing geochronological frameworks for sedimentary deposits (Kohfeld & Harrison 2001; Újvári *et al.* 2010; Kang *et al.* 2015; Stevens *et al.* 2016; Zeeden *et al.* 2018; Lougheed & Obrochta 2019). However, most of these techniques do not consider how much the accumulation rate at a particular depth depends on the depth above it (thickness of the overlying deposit). Hence, in this study, we employed the Bayesian age-depth modelling methodology using the r.bacon code of Blaauw and Christen (2011), version 3.00. For the reconstruction of accumulation records for sediments, r.bacon employs Bayesian statistics by combining numerical age dates with prior information such as accumulation rate and variability (memory). Accumulation rates are controlled using a gamma autoregressive semiparametric model with an arbitrary number of subdivisions along the sediment profile (Blaauw & Christen 2011). Here, we set the resolution to 5 cm, thickness to 20 cm and sample size to 12 000. These settings proved to be appropriate for all investigated LPSs. Prior knowledge about accumulation rates can lead to reduced uncertainty and more realistic modelling (Blaauw & Christen 2011). The priors for accumulation rate and memory (autocorrelation) are modelled by gamma and beta distributions, and the latter defines how much the accumulation rate at a particular depth depends on the depth (thickness) above it. The thickness of the section determines, to some degree, the flexibility of the age-depth model, which is obtained by millions of Markov Chain Monte Carlo (MCMC) iterations (Blaauw & Christen 2011). The used input data (ages, errors and depths) are presented in Data S1.

Following age-depth modelling, we calculated the accumulation rates for each individual LPS based on modelling outputs. For reliable estimation of past dust flux (at least what is preserved) as well as for comparisons of dust archives between different regions, it is necessary to calculate the MARs (Kohfeld & Harrison 2003). The

Table 2. Information on the dating method, dating resolution, dated time period, total number of ages (n), and average number of ages per 10 ka for each published chronology of the investigated LPSs. The total number of ages refers to the number used for the age-depth modelling. Also presented are the references and number of ages for each MIS.

Site	Dating method	Dating resolution (cm)	Dated time period (ka)	n (total)	n (mean per 10 ka)	Reference	MIS 1 (n)	MIS 2 (n)	MIS 3 (n)	MIS 4 (n)	MIS 5 (n)
Bodrogkeresztúr I	pIRIR ₂₉₀	142	28.0–33.5	3	6.0	Bösken <i>et al.</i> (2019)	–	–	3	–	–
Csorgókút I	¹⁴ C	90	17.5–23.5	3	5.0	Sümegei <i>et al.</i> (2007)	–	3	–	–	–
Csorgókút II	¹⁴ C	162	17.5–28.2	4	4.0	Sümegei <i>et al.</i> (2007)	–	4	–	–	–
Debrecen brickyard	¹⁴ C	48	7.1–22.8	7	4.7	Sümegei <i>et al.</i> (2007)	3	4	–	–	–
Dunaszekcső	¹⁴ C	15	23.5–31.3	36	45.0	Újvári <i>et al.</i> (2017)	–	31	5	–	–
Hortobágy, Szálkahalom	¹⁴ C	39	10.0–19.3	5	5.6	Sümegei <i>et al.</i> (2007)	1	4	–	–	–
Kardoskút, Fehér-lake	¹⁴ C	125	8.2–23.3	5	3.3	Sümegei <i>et al.</i> (2007)	2	3	–	–	–
Katymar	¹⁴ C	104	12.6–34.7	10	4.5	Sümegei <i>et al.</i> (2019)	–	8	2	–	–
Lakitelek brickyard	¹⁴ C	118	11.7–30.0	5	2.8	Sümegei <i>et al.</i> (2007)	–	3	–	–	–
Madaras	¹⁴ C	66	11.0–22.0	14	12.7	Sümegei <i>et al.</i> (2020)	–	13	–	–	–
Paks I core	pIRIR ₂₉₀	530	28.2–174.0	4	0.3	Novothny <i>et al.</i> (2020)	–	–	1	–	1
Paks II core	pIRIR ₂₉₀	380	18.6–143.0	5	0.4	Novothny <i>et al.</i> (2020)	–	1	3	–	–
Süttő	IRSL	122	22.1–84.1	9	1.5	Novothny <i>et al.</i> (2011)	–	–	3	2	2
Szeged-Öthalom	¹⁴ C	88	16.0–25.2	5	5.4	Sümegei <i>et al.</i> (2015)	–	5	–	–	–
Tiszaalpári high-bluff	¹⁴ C	179	15.3–27.8	3	2.5	Sümegei <i>et al.</i> (2007)	–	3	–	–	–
Tokaj Kereszt Hill	¹⁴ C	163	0.83–26.9	3	2.8	Sümegei <i>et al.</i> (2007)	1	2	–	–	–
Tokaj Patkó-quarry	¹⁴ C	103	24.9–56.9	9	38.8	Schatz <i>et al.</i> (2012)	–	4	5	–	–
Udvari borehole	¹⁴ C	67	15.7–43.6	6	2.2	Sümegei <i>et al.</i> (2015)	–	4	2	–	–
Batajnica	OSL;	136	12.0–74.0	7	1.1	Avram <i>et al.</i> (2020)	–	2	3	2	–
	pIRIR ₂₉₀										
Belotinac	IRSL	267	22.0–35.0	3	1.7	Basarin <i>et al.</i> (2011)	–	2	–	–	–
Crvenka	OSL;	94	7.7–114.0	12	1.1	Stevens <i>et al.</i> (2011)	1	5	4	1	1
	pIRIR ₂₉₀										
Irig	OSL;	49	7.1–176.9	23	1.4	Perić <i>et al.</i> (2022a, b, c)	2	4	4	4	4
	pIRIR ₂₉₀										
Kisiljevo	OSL	47	0.5–37.0	19	5.1	Perić <i>et al.</i> (2024)	2	8	9	–	–
Nosak	pIRIR ₂₉₀	166	25.1–260.1	14	0.6	Perić <i>et al.</i> (2020)	–	1	3	1	4
Novo Orahovo	OSL	100	15.9–47.8	5	1.6	Marković <i>et al.</i> (2023)	–	4	1	–	–
Orlovat	OSL	100	10.4–91.9	7	0.9	Timar-Gabor <i>et al.</i> (2015)	–	1	1	1	4
Požarevac	pIRIR ₂₉₀	218	21.0–233.9	4	0.2	Perić <i>et al.</i> (2021)	–	1	–	1	–
Stari Slankamen	OSL;	159	6.4–148.0	8	0.5	Schmidt <i>et al.</i> (2010)	1	1	1	2	3
	IRSL										
Stalać	pIRIR ₂₉₀	198	28.5–139.0	6	0.5	Bösken <i>et al.</i> (2017)	–	1	1	1	2
Surduk	¹⁴ C	81	6.4–29.3	12	5.5	Hatté <i>et al.</i> (2013)	1	10	–	–	–
Veliki Surduk	OSL	23	0.4–35.8	14	4.0	Perić <i>et al.</i> (2019)	2	8	4	–	–
Erdut	IRSL	290	19.8–61.5	5	1.2	Galović <i>et al.</i> (2009)	–	–	3	–	–
Vukovar	pIRIR ₂₂₅	129	30.0–145.0	10	4.8	Wacha and Frechen (2011)	–	1	2	2	–
Zmajevac	IRSL	263	17.8–121.9	8	0.8	Galović <i>et al.</i> (2009)	–	3	1	2	2

MARs (in $\text{g m}^{-2} \text{a}^{-1}$) were calculated using the following equation:

$$\text{MAR} = \text{SR} \times \text{BD} \times f_{\text{eol}}, \quad (1)$$

where SR is the sedimentation rate (m a^{-1}), BD is the dry bulk density and f_{eol} is the fraction of the sediment that is of aeolian in origin (Kohfeld & Harrison 2001). Here we assume that the loess is of exclusively aeolian origin, hence we use the value of 1 for the calculation.

The BD values for loess in Europe vary in the literature ranging from 1.65 g cm^{-3} (Frechen *et al.* 2003) to $1.490 \pm 0.008 \text{ g cm}^{-3}$ (Perić *et al.* 2020). In this study, we use the estimate for the Middle Danube sections, which yielded average values of 1.5 g cm^{-3} (Újvári *et al.* 2010). To achieve a more realistic depiction of the MARs, the data underwent smoothing via the loess technique (e.g. Jacoby 2000), incorporating tricube weighting and polynomial regression within the SigmaPlot software version 11.0 (with a sampling proportion of 0.100 and

polynomial degree set to 1). The modelled ages, calculated SR, and unsmoothed and smoothed MARs are shown in Data S1. The corresponding age-depth models for Hungary, Serbia and Croatia are shown in Figs S1–S3, respectively.

Distances for each investigated LPS were calculated via the Near tool, within the Proximity-Analysis toolbox in ArcMap 10.8.2. Input features were longitudes and latitudes of the 34 LPSs and shapefiles (.shp) of the Danube River, Tisza River, loess, loess derivatives, aeolian sand and alluvium. The position, area and boundaries of these geological features were determined according to Lehmkuhl *et al.* (2018). All the features are in the same geographical coordinate system. The six output numbers for each LPS represent the shortest possible distance (straight line) between LPS and shapefiles. If a LPS is located on a loess plateau, the distance to the loess is automatically zero, as the point is on the polygon. The Near tool calculates the shortest distance regardless of the feature type: polylines (rivers), polygons (geology) or points (LPS). No specified radius was set, and the geodetic method was used to obtain the distances in metres, as it includes the curvature of the spheroid. In the case of polygons, the distance is calculated from the edge of the polygon and not from the centre of the polygon. Shapefiles of geological units in this analysis were downloaded from Lehmkuhl *et al.* (2018). It is important to note that it was not possible to determine the accurate position change of the Danube River and its tributaries during the last five MI stages. This limitation arises because river channels are naturally dynamic and can shift significantly over time due to various geological, climatic and environmental factors. The Danube, as a large river in a tectonically active and climatically variable region, will have undergone numerous shifts in its course over glacial–interglacial cycles. However, detailed palaeo-channel data covering each MI stage along the river's entire course are unavailable, and reconstructing past river locations requires extensive stratigraphical and sedimentary records that are rarely continuous or consistent enough to pinpoint precise historical locations for each stage. Therefore, in this study, the Danube's present-day position is used as a reference, providing a practical and consistent baseline for distance calculations. This approach, while a simplification, allows for standardized comparison across LPSs but does not account for past river migrations.

Results

Age-depth models and mass accumulation rates

The Bacon age-depth modelling results are presented in Figs S1–S3. It is noteworthy that the *r.bacon* age-depth models are sensitive to changes in ages with depth in those cases where the dating resolution (<100 cm) was

sufficient to detect such variations (e.g. Dunaszekcső, Hortobágy Szálkahalom, Madaras, Irig, Kisiljevo, Surduk, Veliki Surduk), which resulted in non-linear age-depth functions. For chronologies with lower dating resolution (>100 cm), the age-depth function was evidently more linear (e.g. Bodrogkeresztúr, Kardoskút, Paks, Batajnica, Belotinac, Nosak, Erdut). For 31 investigated sites, 100% of the dates overlap with the age-depth model (95% confidence). The only three exceptions were the Tokaj Kereszt Hill LPS, where only 67% of the dates agree with the *r.bacon* age-depth model, Paks I (75%) and Paks II (80%). The reason for this may be a very high age jump between the uppermost and underlying samples. Such age jumps suggest an erosional event at these sites. Although we attempted to account for possible erosion by including appropriate settings for the modelling process (Blaauw & Christen 2011), the final results could not be improved.

Although, for some sites, the MARs were previously published, the age-depth modelling in these studies was performed using a different method. For the Nosak LPS Perić *et al.* (2020) used the ADmin age-depth model (Zeeden *et al.* 2018) and determined an average MAR of $265 \text{ g m}^{-2} \text{ a}^{-1}$ for the MIS 7–MIS 3 period. Stevens *et al.* (2011) derived their age-depth model for the Crvenka LPS by linearly interpolating between luminescence ages. The accumulation rates were then calculated from the quotient of the difference between two ages and the difference in depth. The mean MAR values for the last glacial cycle at this site range from 138 to $3122 \text{ g m}^{-2} \text{ a}^{-1}$ with a mean of $274 \text{ g m}^{-2} \text{ a}^{-1}$ for the 33–15 ka period. In this study, we also include these LPSs. However, for consistency, we use the same modelling method (*r.bacon*) as for all other sites discussed here. It must be highlighted that for the majority of sites, the chronologies were constructed using relatively few ages (Table 2). Out of the 34 investigated sites, 44% (14 sites) have fewer than two dates per 10 ka available (Table 2), which is why it was not always possible to detect potential hiatuses, erosion events or subtle changes in sedimentation rate. Moreover, for some sites, certain MI stages were only partially dated (e.g. Sümegi *et al.* 2007; Basarin *et al.* 2011; Stevens *et al.* 2011), which further impedes the accuracy of the MAR calculations for that specific period. Finally, there are still only sporadic chronologies available for LPSs beyond MIS 4–MIS 5 (most cover the period of MIS 3–MIS 1). For this reason, we do not include the MAR record for periods beyond MIS 5 in the Discussion. The mean MARs for each investigated LPS in Hungary, Serbia and Croatia are presented in Table 3 and Figs S4–S6, respectively.

The MIS 5 interglacial period is characterized by the lowest MAR values with a mean ($\bar{x}$) value of $152 \pm 24 \text{ g m}^{-2} \text{ a}^{-1}$ (Fig. 2). For this period, the lowest average MARs were observed at the Irig LPS ($53 \text{ g m}^{-2} \text{ a}^{-1}$) while the highest dust accumulation was found for the Zmajevac LPS in Croatia with a value of $288 \text{ g m}^{-2} \text{ a}^{-1}$. During MIS 4 the

Table 3. Mean MAR values (in $\text{g m}^{-2} \text{a}^{-1}$) calculated according to the r.bacon age-depth modelling results for each MIS.

Site	Dating method	MIS 1 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 2 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 3 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 4 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 5 MAR ($\text{g m}^{-2} \text{a}^{-1}$)
Bodrogkeresztúr I	pIRIR ₂₉₀	–	–	732.2	–	–
Csorgókút I	¹⁴ C	–	1244.4	–	–	–
Csorgókút II	¹⁴ C	–	773.4	–	–	–
Debrecen brickyard	¹⁴ C	254.4	327.2	–	–	–
Dunaszekcső	¹⁴ C	–	1921.8	793.5	–	–
Hortobágy, Szálkahalom	¹⁴ C	178.5	207.0	–	–	–
Kardoskút, Fehér-lake	¹⁴ C	394.1	502.4	–	–	–
Katymar	¹⁴ C	–	898.3	346.8	–	–
Lakitelek brickyard	¹⁴ C	–	440.1	–	–	–
Madaras	¹⁴ C	–	1750.0	–	–	–
Paks I core	pIRIR ₂₉₀	–	–	220.0	223.7	272.4
Paks II core	pIRIR ₂₉₀	275.2	275.0	189.6	181.1	154.3
Süttő	IRSL	–	–	309.5	237.3	266.8
Szeged-Öthalom	¹⁴ C	–	515.4	–	–	–
Tiszaalpári high-bluff	¹⁴ C	–	391.2	–	–	–
Tokaj Kereszt Hill	¹⁴ C	246.4	278.8	–	–	–
Tokaj Patkó-quarry	¹⁴ C	–	808.1	684.9	–	–
Udvari borehole	¹⁴ C	–	176.2	154.1	–	–
Batajnica	OSL; pIRIR ₂₉₀	–	201.1	254.6	265.8	–
Belotinac	IRSL	–	724.0	–	–	–
Crvenka	OSL; pIRIR ₂₉₀	179.8	292.7	201.2	163.3	119.5
Irig	OSL; pIRIR ₂₉₀	84.0	132.3	82.8	106.8	52.6
Kisiljevo	OSL	110.4	395.3	501.9	–	–
Nosak	pIRIR ₂₉₀	–	164.5	177.3	166.0	160.6
Novo Orahovo	OSL	–	352.1	193.0	–	–
Orlovat	OSL	–	73.3	76.3	107.7	94.7
Požarevac	pIRIR ₂₉₀	–	49.9	50.0	54.1	60.9
Stari Slankamen	OSL; IRSL	141.2	200.5	185.3	157.5	125.9
Stalać	pIRIR ₂₉₀	–	169.8	161.9	136.1	133.3
Surduk	¹⁴ C	247.9	1095.8	–	–	–
Veliki Surduk	OSL	91.4	194.8	149.8	–	–
Erdut	IRSL	–	–	682.9	–	–
Vukovar	pIRIR ₂₂₅	–	253.5	117.8	114.7	96.8
Zmajevac	IRSL	–	426.1	219.8	256.2	288.1
Total mean		203.3±28.3	507.8±85.5	294.8±48.8	166.9±17.9	152.2±23.5

calculated average MARs were $167 \pm 18 \text{ g m}^{-2} \text{a}^{-1}$ with the highest MARs recorded at the Batajnica LPS in Serbia ($266 \text{ g m}^{-2} \text{a}^{-1}$) and the lowest at the Požarevac LPS ($54 \text{ g m}^{-2} \text{a}^{-1}$) in northeastern Serbia. The MIS 3 period experienced 1.8 times higher MARs than the preceding MIS 4, averaging $295 \pm 49 \text{ g m}^{-2} \text{a}^{-1}$. This is in large disagreement with most studies conducted on loess deposits (e.g. Kohfeld & Harrison 2003) where considerably higher dust fluxes were assumed during the MIS 4 glacial phase than during the following MIS 3 transitional warm period.

The calculated MARs for the LPSs in the Middle Danube Basin are highest during MIS 2, reaching a mean value of $508 \pm 86 \text{ g m}^{-2} \text{a}^{-1}$. The MARs for this period range from $50 \text{ g m}^{-2} \text{a}^{-1}$ (Požarevac LPS, Serbia) to $1922 \text{ g m}^{-2} \text{a}^{-1}$ (Dunaszekcső LPS, Hungary), and are in agreement with most previous studies (Sümegi *et al.* 2007; Újvári *et al.* 2010; Stevens *et al.* 2011; Perić *et al.* 2019), which observed the highest MARs during MIS 2.

Considering the regional context, the highest MARs were calculated for the northern part of the Danube

Basin (Hungary), for all investigated periods (MIS 5–MIS 1), while the lowest were determined for the LPSs situated in the south and southeastern Middle Danube Basin (Serbia). For instance, for MIS 2 the calculated MARs for the Hungarian sites averaged $701 \pm 141 \text{ g m}^{-2} \text{a}^{-1}$ while the Serbian LPSs recorded an average of $311 \pm 81 \text{ g m}^{-2} \text{a}^{-1}$. For the LPSs situated in Croatia, the mean MAR values during the same period were $340 \pm 86 \text{ g m}^{-2} \text{a}^{-1}$. Thus, during all periods the MAR estimates from the northern Middle Danube basin sites are more than twice as high as the ones situated in the south.

Regarding the geomorphological setting, which is considered to have a major effect on the dust accumulation rates at individual LPSs (Kohfeld & Harrison 2003), the MAR values are somewhat variable, also depending on which MIS is investigated. Throughout MIS 5, the MARs are highest at sites located on river terraces, displaying an average value of $154 \pm 60 \text{ g m}^{-2} \text{a}^{-1}$. In MIS 4, the highest MAR values are recorded at loess plateau sites ($\bar{x} = 181 \pm 19 \text{ g m}^{-2} \text{a}^{-1}$). The lowest MARs for MIS 4, according to the geomorphological

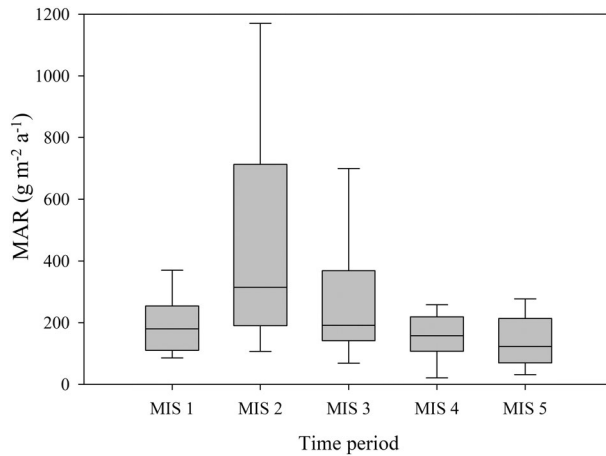


Fig. 2. Box-whisker plots of mean MARs derived from loess profiles located in the Middle Danube Basin covering the MIS 5–MIS 1 period. The grey box represents the interquartile range of the data, the line in the box shows the median and the lines above and below the box show the maximum and minimum values. The error bars are given as 1 standard error of the mean.

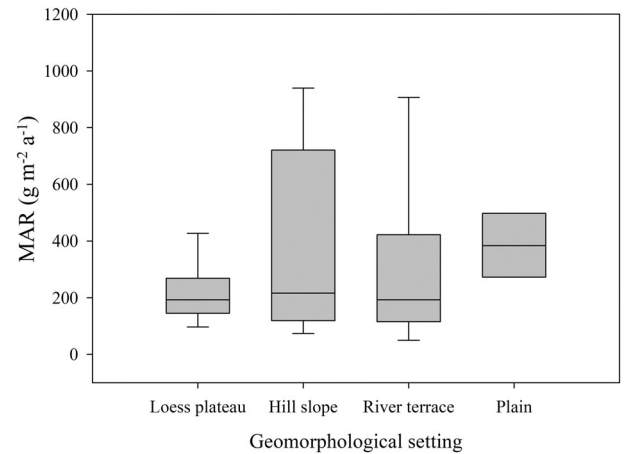


Fig. 3. Box-whisker plots of mean MARs for the Middle Danube Basin LPSs according to the geomorphological setting. The grey box illustrates the interquartile range of the data, with the line inside the box indicating the median, while the lines extending above and below the box mark the maximum and minimum values. The error bars are given as 1 standard error of the mean.

setting, cannot be reliably estimated due to the lack of dated sites for this period. According to the calculated results, the lowest MARs would be found at hill slope sites. However, only one hill slope site (Irig LPS) is dated beyond 50 ka, which is why we do not consider these data representative. During MIS 3 the mass accumulation rates are highest for sites located on hill slopes ($\bar{x} = 467 \pm 143 \text{ g m}^{-2} \text{ a}^{-1}$) ranging from $83 \text{ g m}^{-2} \text{ a}^{-1}$ at the Irig LPS (Serbia) to $732 \text{ g m}^{-2} \text{ a}^{-1}$ at Bodrogkeresztúr I (Hungary). The lowest mean dust flux estimates were observed for loess plateau sites with an average value of $227 \pm 40 \text{ g m}^{-2} \text{ a}^{-1}$. For the following MIS 2 period, the MARs for hill slope and river terrace/bank sites are comparable, averaging 569 ± 181 and $567 \pm 242 \text{ g m}^{-2} \text{ a}^{-1}$, respectively. During this period, the highest MARs were observed at the Dunaszekcső LPS (river bank) with an average value of $1922 \text{ g m}^{-2} \text{ a}^{-1}$ and the lowest at Požarevac LPS (river terrace) where a mean value of $50 \text{ g m}^{-2} \text{ a}^{-1}$ was recorded. The Holocene (MIS 1) is characterized by considerable atmospheric dust activity, with the highest MARs observed at LPSs situated on river terraces ($\bar{x} = 286 \pm 108 \text{ g m}^{-2} \text{ a}^{-1}$) while the lowest values were determined for hill slope sites ($\bar{x} = 165 \pm 81 \text{ g m}^{-2} \text{ a}^{-1}$). The highest dust flux for MIS 1 was determined at the river terrace LPS Kardoskút in Hungary with a mean value of $394 \text{ g m}^{-2} \text{ a}^{-1}$ and the lowest for the hill slope LPS Irig in Serbia with an average MAR of $84 \text{ g m}^{-2} \text{ a}^{-1}$. Generally, over the entire MIS 5–MIS 1 period, the MARs are lowest at LPSs located on loess plateaus ($\bar{x} = 246 \pm 63 \text{ g m}^{-2} \text{ a}^{-1}$), followed by hill slope and river terrace sites with average values of 272 ± 103 and $296 \pm 77 \text{ g m}^{-2} \text{ a}^{-1}$, respectively (Fig. 3). The highest MARs, in this case, would be situated at plain sites with an average value $341 \pm 87 \text{ g m}^{-2} \text{ a}^{-1}$. However, this geomorphological setting is represented

by only three LPSs (Debrecen Alföldi brickyard, Lakitelek brickyard and Szeged-Öthalom) with chronologies covering MIS 2–MIS 1, which is why the calculated data cannot be considered as representative.

In most studies, the proximity to the main silt source is usually only partially addressed as a significant factor for MAR estimates. Here, we calculated the mean values for LPSs located at distances below and above 10 km from the assumed main dust sources: the Danube and Tisza rivers. The LPSs located <10 km from either of the two rivers showed a generally higher dust flux throughout MIS 5–MIS 2 (Table 4). The largest difference was found for MIS 5 where for the sites located <10 km from the Danube River a mean value of $195 \pm 130 \text{ g m}^{-2} \text{ a}^{-1}$ was obtained, while for the >10 km LPSs this value was 2.1 times lower, averaging at $92 \pm 16 \text{ g m}^{-2} \text{ a}^{-1}$. The lowest difference was observed for MIS 2 (1.1 times: <10 km = $548 \pm 196 \text{ g m}^{-2} \text{ a}^{-1}$ and >10 km = $491 \pm 92 \text{ g m}^{-2} \text{ a}^{-1}$). The distance to potential secondary dust sources shows an irregular trend (Table 4). However, we have to highlight that, for the >10 km sites, the MIS 5–MIS 3 periods are represented by only one site, which can considerably affect the mean MAR value. The proximity to loess derivatives does not appear to have a major effect on the sedimentation rate. The calculated mean MAR values (MIS 5–MIS 1) for <10 km totals $292 \pm 75 \text{ g m}^{-2} \text{ a}^{-1}$, whereas for the >10 km sites the MARs average $284 \pm 74 \text{ g m}^{-2} \text{ a}^{-1}$, making the difference negligible. A major effect on the sedimentation rates seems to be the proximity to aeolian sand sources and alluvium. The sites with a distance <10 km to aeolian sands average $327 \pm 70 \text{ g m}^{-2} \text{ a}^{-1}$, which is higher by a factor of 1.5 when compared to the value of $218 \pm 54 \text{ g m}^{-2} \text{ a}^{-1}$ for the >10 km sites over the MIS 5–MIS 1 period. The largest difference, however,

Table 4. Mean MAR values calculated according to the distance of LPS to the silt sources. The MARs ($\text{g m}^{-2} \text{a}^{-1}$) are shown for each MIS and present the average value for all investigated sites.

Dust source	Distance	MIS 1 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 2 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 3 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 4 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	MIS 5 MAR ($\text{g m}^{-2} \text{a}^{-1}$)	Mean MIS 1–MIS 5 MAR ($\text{g m}^{-2} \text{a}^{-1}$)
Danube	<10 km	194±40	548±196	332±68	200±19	195±30	294±69
River	>10 km	204±40	491±92	258±71	114±18	92±16	232±71
Tisza River	<10 km	182±39	594±117	438±157	158 ¹	126 ¹	299±92
	>10 km	211±40	442±124	258±51	168±19	155±26	247±52
Loess	<10 km	202±32	555±101	359±73	184±25	159±40	292±75
derivates	>10 km	275 ¹	575±249	202±28	185±29	185±41	284±74
Aeolian	<10 km	256±45	596±113	368±74	202±17	214±32	327±73
sand	>10 km	154±25	420±128	244±63	151±23	121±26	218±54
Alluvium	<10 km	185±33	576±107	325±57	173±20	165±26	285±78
	>10 km	218±52	302±52	159±38	135±28	86±33	180±37

¹MAR data available for only one site.

was observed when considering the proximity of LPSs to alluvium sources. In this case, the average MARs for <10 km sites ($\bar{x} = 285 \pm 78 \text{ g m}^{-2} \text{a}^{-1}$) are higher by a factor of 1.6 when compared to the >10 km sites ($\bar{x} = 180 \pm 37 \text{ g m}^{-2} \text{a}^{-1}$).

Discussion

Regional context of MAR variability

Current global atmospheric dust activity and past climate change models are mostly based on data from ice-cores (e.g. Augustin 2004) and marine sediment cores (e.g. Lisiecki & Raymo 2005). These records suggest major increases in atmospheric dust load during cold glacial and stadial periods, as opposed to warmer interglacials and interstadials (Kohfeld & Harrison 2003; Ruth *et al.* 2003; Lambert *et al.* 2008). However, the terrestrial sedimentary systems react differently to climate changes, which makes a correlation between ice and marine core records with terrestrial sediment (loess) archives somewhat challenging. For this correlation to be accurate, we need to account for varying sediment accumulation rates and partial preservation, and for this purpose, accurate and robust chronologies are required. Additionally, we need to consider the potential lag in terrestrial ecosystem response to past climate change. This includes geomorphological processes, vegetation changes and soil development. As evidenced by the high-resolution dust record of central Europe, the reforestation in that region was significantly delayed during the Lateglacial period (e.g. Street *et al.* 2001). For instance, over longer time scales, Sier *et al.* (2015) reported a 5000-year diachronic offset in the onset of Eemian interglacial conditions between southern and northern Europe, determined through the correlation of high-resolution terrestrial pollen records and marine archives. Furthermore, the assessment of mass accumulation in the Middle Danube Basin should consider potential biases arising from the sampling position near

upper and lower soil boundaries. This mixing of depositional contexts could lead to inaccuracies in the MAR results, as similar dynamics were observed during the transition from MIS 2 to MIS 1, where dust accumulation coincided with soil formation (An *et al.* 1991; Liu & Ding 1998; McGee *et al.* 2010).

Thus far, multiple studies have been able to successfully compare the ice and marine core records with terrestrial sediment (loess) archives (Kohfeld & Harrison 2003; Marković *et al.* 2014, 2015; Stevens *et al.* 2016; Rousseau *et al.* 2018). Indeed, a trend of high atmospheric dust flux was noted during the drier, colder phases, and low during the more humid, warmer periods (Újvári *et al.* 2017), which agrees with the dust flux models based on the ice and deep sea core archives. The accumulation and preservation of loess and palaeosols are influenced by the interaction of various factors: topography (including slope, aspect, and curvature), climate (such as temperature, wind strength, and direction), and sediment/soil properties (like grain size, calcification, and decalcification) within a particular vegetation cover (Lehmkuhl *et al.* 2021). The characteristics of LPSs in the Middle Danube Basin suggest a palaeoclimatic gradient trending towards warmer and drier conditions from northwest to southeast (Sümegei & Krollop 2002). The drier climate conditions in this region contributed to better preservation and subsequently more complete LPSs and higher sedimentation rates (Marković *et al.* 2008, 2015; Antoine *et al.* 2009; Bokhorst *et al.* 2009; Újvári *et al.* 2010, 2017; Sümegei *et al.* 2013). The LPSs investigated in this study are all positioned outside the permafrost zone: five sites are located at the boundary of the discontinuous permafrost and seasonal frost zone (e.g. Bodrogekresztúr I, Csorgókút I, Csorgókút II), one LPS is found in the discontinuous permafrost zone, while the rest (28 LPS) are situated in the generally warmer, seasonal frost zone (Fig. 1). In these climatic conditions, biological loess crusts and mats were acting as dust traps and possibly contributed to the loessification process, converting the

semi-continuous dust accumulation into stable LPSs (Svirčev *et al.* 2013, 2019). Along with vegetation from semi-desert to steppe environments, these biocrusts provided effective protection for LPSs against erosion and deflation, resulting in plateau deposits that document Pleistocene environmental history from at least the Late Lower Pleistocene (Lehmkuhl *et al.* 2021). Additionally, in major parts of the investigated area, especially towards the southeasternmost limit of the Middle Danube Basin, the Late Pleistocene climate was generally dry, and no evidence of periglacial processes was observed. Contrary to some other European loess regions where periglacial environments were prevalent (e.g. central Europe), our study area was the least exposed to postdepositional processes (e.g. Marković *et al.* 2008, 2018; Stevens *et al.* 2011; Vandenbergh *et al.* 2014).

The regional distribution of MARs in the Middle Danube Basin for each investigated MIS is presented in Fig. 4. The MAR values display a mean value of $317 \pm 35 \text{ g m}^{-2} \text{ a}^{-1}$ for the investigated MIS 5–MIS 1 period, ranging from $50 \text{ g m}^{-2} \text{ a}^{-1}$ at the Požarevac LPS in Serbia (MIS 2) to $1922 \text{ g m}^{-2} \text{ a}^{-1}$ at the Dunaszekcső LPS in Hungary (MIS 2). The calculated mean MARs for MIS 2 ($508 \pm 86 \text{ g m}^{-2} \text{ a}^{-1}$) show a fairly good correspondence (within uncertainties) with the published results of Újvári *et al.* (2010), where an average MAR value of $417 \pm 50 \text{ g m}^{-2} \text{ a}^{-1}$ was determined for the Carpathian Basin, albeit the study used a different modelling method (linear interpolation).

The calculated MARs for the Middle Danube LPS, according to the published chronologies, only partially confirm that dust accumulation in this region was accelerated during the glacial periods compared to interglacials. Our results suggest that MARs during MIS 5 ($\bar{x} = 152 \pm 24 \text{ g m}^{-2} \text{ a}^{-1}$) were only slightly lower when compared to MIS 4 ($\bar{x} = 167 \pm 18 \text{ g m}^{-2} \text{ a}^{-1}$), suggesting that a reduced, but near-continuous dust sedimentation may have occurred in the region over the last interglacial. Regarding the MAR estimates for MIS 3, these exhibit 1.8 times higher values than for MIS 4, averaging at $295 \pm 49 \text{ g m}^{-2} \text{ a}^{-1}$. Regionally, this tendency is also observable for the LPSs in Hungary and Serbia. The exception is Croatia, where the mean MAR values for MIS 4 ($\bar{x} = 193 \pm 96 \text{ g m}^{-2} \text{ a}^{-1}$) slightly exceed those for MIS 3 ($\bar{x} = 186 \pm 71 \text{ g m}^{-2} \text{ a}^{-1}$). Still, here we have to consider the fact that Croatia is represented by only three LPSs, where one site encompasses ages only for MIS 3 (Erdut), while for Vukovar and Zmajevac MIS 4–MIS 2 and MIS 5–MIS 2 chronologies are available. Yet, only Vukovar was dated in a dating resolution (4.8 dates per 10 ka; Table 2), which is capable of providing a more detailed insight into the MAR variability during the individual MI stages. Such a sample size is not sufficient to draw conclusions about the dust deposition trend for an entire region, which is why we find these results debatable. Additionally, as already stated, it is important to consider the spatial context of the sampled sediments,

particularly their proximity to soil boundaries, which may introduce bias in MAR estimates. The Eemian interglacial (MIS 5e) for example is characterized by significant climatic transitions and associated pedogenic processes. Evidence from the transition between MIS 2 and MIS 1 indicates that in the Middle Danube Basin, dust accumulation dominated while soil formation was also underway. Could similar conditions have occurred during MIS 5, with soils developing from MIS 6 sediments and yielding mixed ages? This could overestimate MAR if soil ages do not accurately reflect loess deposition, highlighting the need to examine sampling positions and sediment characteristics relative to their chronological assignments. One site, where all the MAR estimates appear to fit the generally accepted model of accelerated dust accumulation during glacial periods and decreased during interglacial periods is the Irig LPS in northeastern Serbia (Fig. 1, Tables 1, 3), which is located on a gentle hill slope. However, considering the average values for all investigated LPSs, the MAR pattern covering the MIS 5–MIS 3 period in the Middle Danube Basin appears to contradict the generally accepted dust flux model, provided that the MAR estimates are correct and not affected by dating resolution (i.e. fewer ages available for MIS 4 than for MIS 3). If correct, this controversy is especially evident in comparison with the MAR models for the Chinese Loess Plateau (Kohfeld & Harrison 2003), where a very consistent MAR trend was observed for the entire MIS 3–MIS 5 period: during the colder, glacial (stadial) phases accelerated dust deposition was observed, while for the interglacial (interstadial) cycles a decrease in dust accumulation was found (Kohfeld & Harrison 2003). However, there are multiple causes that could have affected the accuracy of the observed MAR pattern in the Middle Danube Basin, such as low dating resolution, age distribution, or biases in the dating method. In addition to these factors, erosional hiatuses and re-deposition processes such as slope-driven movement or sheetwash could also significantly impact MAR estimates. Erosional hiatuses, if present, would result in missing sediment layers and thus yield artificially low MARs over affected intervals. Conversely, re-deposition of sediment from upslope sources could inflate MAR estimates in certain layers by adding material from older deposits. This is particularly relevant in LPSs on hill slopes or near areas prone to slope instability, where increased sediment reworking can alter primary dust accumulation signals.

One of the most important factors in identifying dust flux patterns over a certain time period is a sufficient number of reliable ages. Out of the 34 investigated LPS, 20 sites (59%) were dated in a resolution lower than 100 cm, while only 15 sites (44%) have more than three ages per 10 ka available (Table 2). This issue is not so pronounced if the chronology covers only one MIS as in the case of Bodrogheresztúr I, Csörgökút I, Csörgökút

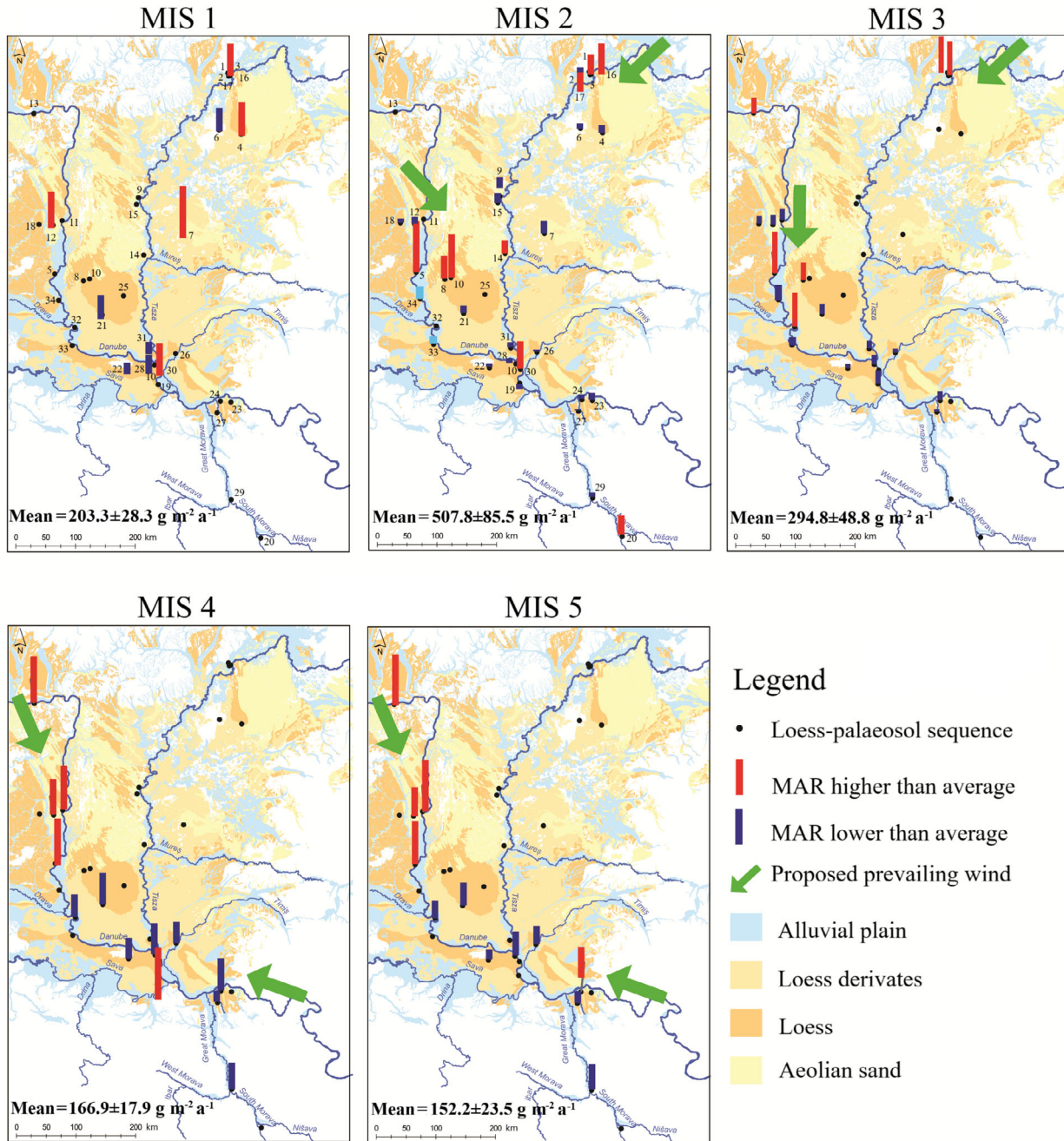


Fig. 4. Maps of the Middle Danube Basin displaying mean MARs for the last five Marine Isotope Stages (MIS 1–5). Each panel shows the mean MAR ($\text{g m}^{-2} \text{ a}^{-1}$) with uncertainty, along with site-specific MAR values for individual loess-palaeosol sequences. The colour-coded columns indicate sites where MARs are higher (red) or lower (blue) than average, and green arrows represent the proposed prevailing wind directions. The background map highlights geomorphological features, including alluvial plains, loess derivatives, loess deposits, and aeolian sand, as specified in the legend.

II, Lakitelek brickyard, Szeged-Óthalom or Tiszaalpári high-bluff. However, the problem becomes evident if the chronology spans over an extended period, covering multiple MISs. This makes the detection of MAR variability for a certain time very difficult because, in this instance, the age-depth modelling provides values

averaged over a longer period, thus yielding unrealistic values for an MIS with few or no ages at all. This is most noticeable in the case of the Požarevac LPS where the published chronology covers the MIS 7–MIS 2 period, but with only one age available for MIS 2 and one age for MIS 4. It is evident that the calculated MARs for MIS

5–MIS 2 display very low variability ($49.9\text{--}60.9\text{ g m}^{-2}\text{ a}^{-1}$). This suggests that the calculated MARs represent the average dust flux over the entire MIS 5–MIS 2 period, rather than the actual values for the individual stages. Likewise, the age distribution would affect the modelling results in a similar manner, even when the average dating resolution is considered satisfactory. For example, the dating resolution for the Orlovat LPS is 100 cm (0.9 ages per 10 ka); however, only MIS 5 is represented by four ages, while for MIS 4–MIS 2 one age per stage is available. Here, the highest MARs are observed during MIS 4 ($107.7\text{ g m}^{-2}\text{ a}^{-1}$) while the lowest values were noted during MIS 2 ($73.3\text{ g m}^{-2}\text{ a}^{-1}$). It is noticeable that MIS 3 displays slightly higher MARs than MIS 2 with an average of $76.3\text{ g m}^{-2}\text{ a}^{-1}$. Still, MIS 5 shows a higher accumulation rate than MIS 2 averaging $94.7\text{ g m}^{-2}\text{ a}^{-1}$, which contradicts the general dust deposition model. It is conceivable that in this case, the r.bacon software, in the absence of a sufficient number of data points (ages), constructed a rather averaged, linear age-depth model, which resulted in MARs with unrealistically constant values for the entire investigated time period. Additionally, chronologies with a low dating resolution are not always capable of detecting erosional events, sedimentation breaks or slight fluctuations, which can significantly hinder the accuracy of dust flux estimates. One other element that can hamper the accuracy of the calculated MARs is biases in the dating method for different time periods as the average accumulation rates can vary significantly depending on the employed dating technique (Kohfeld & Harrison 2003). In particular, there is the question of which dating method is more reliable for dating loess sequences. For instance, the mean MARs during MIS 2 when using ^{14}C dating range from 207 to $1922\text{ g m}^{-2}\text{ a}^{-1}$, with an average of $755\pm140\text{ g m}^{-2}\text{ a}^{-1}$, while for MIS 1, the MARs vary from 179 to $394\text{ g m}^{-2}\text{ a}^{-1}$, with a mean of $264\pm35\text{ g m}^{-2}\text{ a}^{-1}$. Luminescence age-generated MARs for MIS 2 are in the range of $50\text{--}426\text{ g m}^{-2}\text{ a}^{-1}$ for MIS 2 ($\bar{x} = 260\pm43\text{ g m}^{-2}\text{ a}^{-1}$), whereas for MIS 1 the MARs range from 84 to $275\text{ g m}^{-2}\text{ a}^{-1}$ ($\bar{x} = 147\pm29\text{ g m}^{-2}\text{ a}^{-1}$). It is notable that in both cases the ^{14}C chronologies suggest 2.9 and 1.8 times higher MAR values for MIS 2 and MIS 1, respectively, than the luminescence dates. The calculated MIS 2/MIS 1 ratio in this case is 1.8 for the luminescence and 2.9 for the ^{14}C chronologies. However, this comparison cannot be considered a viable analysis of the difference amongst MAR values generated using diverse dating methods. One major issue is that the ^{14}C and luminescence chronologies of most investigated LPSs in this study are not randomly sampled data sets (Kohfeld 2000). For example, 63% of the luminescence dated sites are loess plateaus but only for 26% of these sites is it possible to obtain MAR values for MIS 1. In the case of ^{14}C , the most represented sites are hill slopes with 33%, but for only one site (7%) is it possible to derive MARs for MIS 1. Consequently, if MAR values obtained for hill slopes

differ from the MARs calculated for other types of loess (e.g. loess plateaus) these estimates would be biased. One other specific problem can also arise due to larger error limits when using luminescence chronologies, particularly if certain ages are near the end or at the onset of a Stage. For example, at the Paks II LPS, the youngest age was $18.6\pm1.2\text{ ka}$. This allowed the r.bacon software to partially include MIS 1 in the age model, although no actual date from this period is available (excluding uncertainty). Therefore, the accuracy of MARs calculated for MIS 1 in this case are, to a certain degree, questionable.

Geomorphological setting as a factor in MAR estimates

It is expected that the continuity and thickness of dust deposition are strongly influenced by the geomorphological situation. Plateau and plain sites are considered to be relatively stable settings and are therefore more likely to preserve a more continuous accumulation record and be less prone to postdepositional reworking (Frechen *et al.* 2003). Hill slope sites on the other hand, are deemed to be more susceptible to reworking and may contain sedimentary hiatuses (e.g. Kohfeld & Harrison 2003). There are eight hill slope sites investigated in this study, of which only for one (Irig LPS) a chronology covering the entire MIS 5–MIS 1 period is available. All other sites cover only a single MIS (e.g. Bodrogekresztúr I, Csörgökút I, Csörgökút II) or maximally two Stages (e.g. Tokaj Kereszt Hill, Udvari). This makes it problematic to determine the level of preservation according to the published chronology as there is no possibility of observing large age jumps (indicator of erosional processes) or age inversions (indicator of postdepositional processes). Still, at Irig LPS, where the high-resolution OSL/pIRIR chronology covers the entire MIS 5–MIS 1 period, no jumps or inversions (outside uncertainties) could be observed, suggesting a high level of loess preservation. The possible causes for this can largely be ascribed to a number of site-specific factors, such as topography, local palaeoclimate or physical properties of the sediment (Lehmkuhl *et al.* 2018). Indeed, the palaeoclimatic conditions at the southeastern slopes of the Fruška Gora were generally drier than the rest of the Carpathian Basin (Marković *et al.* 2007, 2008; Vandenberghe *et al.* 2014), whilst not affected by permafrost dynamics. Additionally, the Irig LPS is located on a gentle slope, and away from potentially transient local influences of the Danube River, which resulted in relatively low accumulation rates. These factors could have largely prevented the degradation of the Irig LPS.

The highest mean MARs for the Middle Danube Basin over the entire MIS 5–MIS 1 period are observed at hill slope sites with $\bar{x} = 402\pm90\text{ g m}^{-2}\text{ a}^{-1}$, which is a value 1.5 times greater than for loess plateaus ($\bar{x} = 268\pm41\text{ g m}^{-2}\text{ a}^{-1}$). The average hill slope MAR

values exceed also those at river terraces ($353 \pm 106 \text{ g m}^{-2} \text{ a}^{-1}$) by a factor of 1.1. This is rather surprising as it is considered that LPSs located on river terraces usually exhibit the highest accumulation rates because they are positioned in the immediate vicinity of local dust sources. These results suggest a possible bias in the MAR estimates. Out of all investigated LPSs (34 sites) only 24% are hill slope sites. However, for 25% (Csorgókút I and Csorgókút II) of hill slope LPS, MAR estimates are available only for MIS 2 while for another 25% of the LPSs the MARs were obtainable solely for MIS 3 (Bodrogeresztúr I and Erdut). For one site both MIS 2 and MIS 1 MARs could be calculated (Tokaj Kereszt Hill) with two other sites containing both MIS 3 and MIS 2 MAR estimates (Tokaj Patkó-quarry and Udvari borehole). Only for the Irig site is a chronology covering the entire MIS 5–MIS 1 period available, for which the calculated mean MAR is $94 \pm 9 \text{ g m}^{-2} \text{ a}^{-1}$. This is a significantly lower value than when all the other sites are included. It is obvious that the average MAR value for the hill slope sites is biased as it is averaged with the inclusion of sites where only partial chronologies are available. If the hill slope sites are excluded, the MAR pattern according to the geomorphological setting does not become more consistent: the highest MARs are observed on plain sites ($\bar{x} = 384 \pm 58 \text{ g m}^{-2} \text{ a}^{-1}$) followed by river terrace sites, while the lowest accumulation rates were determined for the plateau sites. This is somewhat surprising as river terrace LPSs are situated close to local riverine sources of dust, which is why, at such sites, the rates of dust accumulation are usually significantly higher during all MIS and particularly during glacial periods (Kohfeld 2000). Still, given the fact that plain sites are represented by only three LPSs (Debrecen, Lakitelek brickyard and Szeged-Öthalom) and the chronologies cover only MIS 2–MIS 1, we cannot consider the calculated MARs as representative enough to make a significant comparison over the entire MIS 5–MIS 1 period. However, when solely the MARs for MIS 2 are compared, plain sites display lower values than plateaus with $\bar{x} = 428 \pm 55$ and $\bar{x} = 491 \pm 145 \text{ g m}^{-2} \text{ a}^{-1}$, respectively. It is obvious that the age distribution plays a significant role when comparing MAR estimates and that the interpretation according to the available results would largely be biased.

Influence of proximity of immediate and secondary immediate dust sources on MAR

The proximity to the immediate dust source (usually major rivers and their tributaries) is frequently neglected when investigating atmospheric dust activity trends, although it is evident that this factor plays a major role in the magnitude of dust accumulation. As already stated, the LPSs located <10 km from the Danube and Tisza rivers generally display higher MAR values than the more distant sites throughout the MIS 5–MIS 2 periods

(Table 3). This is especially evident in the case of the Danube River. For example, during MIS 5, sites with a <10 km distance to the Danube average $195 \pm 30 \text{ g m}^{-2} \text{ a}^{-1}$ as opposed to $92 \pm 16 \text{ g m}^{-2} \text{ a}^{-1}$ for the >10 km sites. Similar trends are also observed for all other stages. For the Tisza River, the most pronounced difference is observed during MIS 3 where sites with a <10 km distance display a mean value of $438 \pm 157 \text{ g m}^{-2} \text{ a}^{-1}$ while the >10 km sites have average MARs of $258 \pm 51 \text{ g m}^{-2} \text{ a}^{-1}$. Unfortunately for MIS 4 and MIS 5, the <10 km LPSs are represented by only one site, which hampers a more detailed analysis. Apparently, shorter dust transportation distances usually result in higher accumulation rates, regardless of the geomorphological setting of the LPS. It is possible that during intensified atmospheric activity (i.e. stronger palaeowinds), both the Danube and the Tisza rivers acted as immediate dust sources, which resulted in higher dust accumulation rates at LPSs located in the vicinity of these rivers. However, such proximity of the LPSs to the immediate dust sources (such as large river systems) most likely hinders a realistic interpretation of the atmospheric dust activity during the investigated time period (Perić *et al.* 2022b, c). Specifically, loess deposits at nearby locations may have been significantly affected by reorganization of the river networks associated with the Danube and its tributaries, as well as changes in sediment load and discharge (Marković *et al.* 2015). Hence, these factors should always be considered when using LPS accumulation rates for the interpretation of atmospheric dust variations.

The contribution of the potential secondary immediate dust sources to the sedimentation rates of an LPS is often difficult to decipher when observing the mean MARs for the single MIS periods. However, when considering the entire MIS 5–MIS 1, the average MARs for all <10 km sites are higher when compared to the >10 km sites (Table 3). The aeolian origin of loess is usually unequivocal, but the interpretation is complicated because loess deposits are adjacent to dust source regions and can serve as both dust sources and sinks. Loess deposits may also contain a mixture of aeolian material from local aeolian reworking and recycling (Kapp *et al.* 2015; Licht *et al.* 2016; Stevens *et al.* 2018), and hemispheric material transported for long distances. Marine cores generally contain finer aeolian material compared to loess, reflecting the importance of local sources and reworking for loess. Recently, Marković *et al.* (2023) highlighted environmental conditions in a dust source area as an important factor in loess formation. Besides prevailing winds, local geomorphology, hydro-climatic conditions and vegetation cover at loess deposition sites, proximity of the dust source area, as well as environmental dynamics of the source area (intensity of the dust emission and grain-size composition) are also crucial. One of the most important factors that it is not possible to fully address at present is the fact

that the distance of dust sources to loess sites likely varied during different MI stages, influencing dust accumulation rates and loess formation. Factors contributing to this variability include river system dynamics (changes in river networks affected sediment availability and transport), landscape evolution, climate-driven changes, vegetation cover fluctuations, wind patterns, permafrost dynamics, and human activities. These factors highlight the complexity of interpreting atmospheric dust activity trends based solely on loess accumulation rates. A comprehensive multi-proxy approach integrating geomorphological, climatological and sedimentological data is necessary to accurately reconstruct and fully understand past dust source dynamics. With the exception of information related to the local vegetation on palaeotopographic surfaces of investigated LPSs (Marković *et al.* 2007, 2008, 2018, 2021; Zech *et al.* 2013; Sümeği *et al.* 2016; Radaković *et al.* 2023), the pieces of this spatial and temporal puzzle are still relatively unknown.

¹⁴C or luminescence chronologies – which are more suitable for MAR estimates?

The chronological framework is obviously critical for the comparison of synchronously deposited loess and the MAR calculation. The most used numerical dating methods in loess research for the last interglacial/glacial cycle are luminescence (OSL, IRSL, pIRIR) and radiocarbon (¹⁴C) dating. Evidently, these methods use different concepts and techniques for age determination. While ¹⁴C determines the age of organic matter (in loess sequences usually snail shells, bones or charcoal), OSL measures the time elapsed since the last exposure of mineral grains (e.g. quartz, feldspar) to sunlight, which makes this method particularly suitable for dating aeolian sediments. Although very different in their basic concept, these two dating methods are considered to be able to provide reliable age estimates for LPSs, thus determining the time of deposition of a particular loess layer.

Nevertheless, there are certainly advantages and disadvantages associated with both methods. Although considered to be more precise (low uncertainty), ¹⁴C is able to date samples only up to ~50 ka, which makes it less usable for deposits beyond MIS 3 (Aitken 2014). There is also very often the issue of sample contamination by younger organic matter and also uncertainties in the ¹⁴C age calibration, which can influence the reliability of the generated ages (Mellars 2006). In the case of luminescence dating, issues like incomplete bleaching, postdepositional mixing, erosional events (which can reset the signal) or improper anomalous fading correction (in the case of feldspar) can significantly impact the accuracy of the constructed chronologies. If not addressed, the issues mentioned will ultimately result in erroneous chronologies, consequently rendering the associated interpretations inaccurate.

As already mentioned, one major issue that can arise when dating feldspar grains using luminescence dating is tunnelling (anomalous fading). Anomalous fading in feldspars is marked by a reduction in luminescence intensity observed during optical or thermal excitation, which occurs as the delay time between irradiation and measurement increases. This phenomenon can be described by logarithmic decay expressed as a percentage of signal loss per decade (Aitken 1998), where a decade represents a factor of 10 increase in delay time since irradiation (Huntley & Lamothe 2001; Lamothe *et al.* 2003). Methods where feldspar measurements are conducted at high temperature, such as pIRIR₂₂₅ or pIRIR₂₉₀, are considered to be less susceptible to anomalous fading (Thiel *et al.* 2011; Buylaert *et al.* 2012, 2015; Roberts 2012). However, the IRSL technique, which usually measures the luminescence signal at temperatures as low as 50 °C, is typically prone to anomalous fading. Conversely, erroneous fading correction can be so pronounced that it can result in a significant age over or underestimation of feldspar samples. For example, a g_{2days} value (normalized measurement delay time of 2 days after irradiation) of ~5% per decade would lead to a fading correction of more than 50% on younger samples (Huntley & Lamothe 2001). This would obviously present a significant discrepancy and if not properly corrected for, lead to erroneous chronologies and subsequently inaccurate MAR estimates. Compared to the low-temperature IRSL, even though less susceptible to anomalous fading, the high-temperature pIRIR signals are known to bleach more slowly and retain residual doses of a few Gy or more. This limits its application to dating relatively young samples and may cause problems for correcting residual dose/signal. If not accounted for, this phenomenon can, in some cases, result in age overestimations of >10 ka (Buylaert *et al.* 2011, 2012; Murray *et al.* 2012).

In this investigation, four sites were dated using solely IRSL (Süttő, Belotinac, Erdut and Zmajevac) and one in combination with quartz OSL (Stari Slankamen). At the Süttő LPS, a fading rate of 1.4–2.5% per decade was determined, which led to an age correction of 20%. The corrected ages showed a very good agreement with the available AAR and radiocarbon dates; however, the fading corrected IRSL ages from the penultimate glacial loess displayed a slight underestimation of the supposed real deposition age. In this case, it was advised that the IRSL data for this period should be interpreted as minimum ages (Novothy *et al.* 2011). For the Belotinac site, fading rates in the range 3.5–3.7% per decade were calculated, which resulted in a ~30% age correction (Basarin *et al.* 2011). Here, the authors highlighted that the luminescence ages are comparable with the results from similar loess sections in the Vojvodina region such as Surduk, Stari Slankamen and Crvenka (Fuchs *et al.* 2008; Bokhorst *et al.* 2009; Schmidt *et al.* 2010; Stevens *et al.* 2011). This suggests that the fading

correction for this site was performed accurately and the published ages should be considered reliable. For the Erdut and Zmajevac sites, there are no details available on the fading measurements (Galović *et al.* 2009). It is therefore unclear whether any fading measurements or corrections were performed. Any issues with the used protocols and age calculations that were not detected and properly addressed may have resulted in inaccurate MAR estimates at these two sites. However, given the fact that the MAR distribution at the Zmajevac LPS mostly agrees with the general dust deposition model (higher accumulation rates during glacial cycles, lower during interglacial periods), it is conceivable that the published chronologies are reliable. The Stari Slankamen LPS was dated using combined quartz OSL and polymineral IRSL. The calculated mean fading rate was 3.24% (for the polymineral IRSL) and the ages were corrected accordingly (Schmidt *et al.* 2010). Their stratigraphical model was found to be in agreement with the results of amino acid racemization (AAR) chronologies for different loess sections in the Vojvodina region (Marković *et al.* 2007, 2008, 2009), implying that the age calculations were conducted correctly.

Although diverse in their basic concept, ^{14}C and OSL are very often used in combination when constructing numerical chronologies (e.g. Almond *et al.* 2007; Zhao *et al.* 2007; Újvári *et al.* 2014; Song *et al.* 2015; Sümeği *et al.* 2022) or in MAR investigations (Frechen *et al.* 2003; Kohfeld & Harrison 2003; Újvári *et al.* 2010). Moreover, luminescence dating very often complements ^{14}C dating in the entire age range over which the latter is applicable: usually up to 40 ka. Multiple conducted studies confirmed that, in many cases, the ^{14}C and OSL ages display a very good to excellent correspondence (Prescott & Robertson 1997). This is especially true for investigations conducted on European sediments where up to an age of 15 ka, the discrepancies were negligible (e.g. Smith *et al.* 1990; Stokes & Gaylord 1993; Selsing & Mejdahl 1994; Fischer *et al.* 2021; Sümeği *et al.* 2022).

Given that no LPS was dated beyond MIS 3 using ^{14}C , and no major issues in the measurements were reported, we can conclude that the published radiocarbon ages are reliable (within uncertainties). As for the luminescence chronologies, it is expected that the calculated ages have larger uncertainties than do the ^{14}C dates, which would yield somewhat less precise MAR estimates. This is especially noticeable when comparing the confidence ranges of the ^{14}C and luminescence modelled ages (Data S1, Figs S1–S3). In all cases, the range between the minimum and maximum modelled ages at a certain depth was considerably smaller for ^{14}C ages than for the luminescence chronologies. The mean confidence ranges for the luminescence sites vary between 4361 years for the Kisiljevo LPS, to 41 427 years for the Požarevac LPS. The calculated mean value for luminescence dated sites is $17\,129 \pm 2461$ years. For the ^{14}C chronologies, these

values range from 605 years for the Madaras LPS, to 7233 years for the Tokaj Kereszt Hill site ($\bar{x} = 2698 \pm 502$ years). It is notable that the mean confidence ranges are higher for the luminescence chronologies than for the ^{14}C ones by a factor of 6. This suggests that the mean ^{14}C ages (used for MAR calculations), would be more precise, albeit only for the MIS 3–MIS 1 period. However, it is debatable whether precision in this case also implies accuracy.

At this time, it is difficult to determine which method provides more reliable chronologies and thus more accurate MAR estimates, given the vast differences in the applied methodologies. On one hand, ^{14}C does not directly date the time of deposition of the sediment but rather the organic material embedded in a certain loess layer, which can lead to misinterpretations. Additionally, this method has a relatively limited upper dating limit of ~ 50 ka. On the other hand, luminescence dating, although it does determine the deposition time of the sediment and is able to, in some cases, measure the age of a sample up to 1 Ma, exhibits larger uncertainties and requires the consideration of multiple factors (e.g. fading correction, residual dose, partial bleaching), which can result in an age over or underestimation.

Not a single LPS investigated in this study was dated using both ^{14}C and luminescence dating, making a comparison and a detailed analysis implausible. Furthermore, at this time, no method is available for determining the ‘true’ age of a sediment sample with which the ^{14}C and OSL dates could be compared, which makes a discussion on the relative suitability of a particular dating technique redundant. Presently, we have to trust the authors of the original chronologies that all factors have been accounted for and the published ages are reliable. Nevertheless, detailed examinations and scepticism are advised when using a chronology that appears to be problematic or where no details on the performed measurements and calculations are available.

Conclusions

Published geochronological data of 34 LPSs from the Middle Danube Basin (Hungary, Croatia and Serbia) were compiled and modelled using *r.bacon*, after which MARs were calculated for each individual site, covering the MIS 5 MIS 1 period. The average MARs presented in this study (Figs 2, 4) generally agree with previous investigations conducted in this region (Újvári *et al.* 2010). The highest MARs were identified at LPSs located closer to the two major silt sources (the Danube and Tisza rivers), where the average values are from 10% (MIS 2) to 50% (MIS 5) higher than at more distant sites. This mostly agrees with previous studies where main dust deposition hot spots were identified along or close to the River Danube (e.g. Újvári *et al.* 2010). Although the vicinity of the LPSs to potential secondary dust sources displays an irregular trend, areas covered with

aeolian sands and alluvium appear to make a significant contribution to the MAR values of the investigated LPSs, displaying 1.5–1.7 times higher values than the more distant sites.

While the dust accumulation pattern in the Middle Danube Basin covering the MIS 3–MIS 1 period (last ~57 ka) is consistent with the generally accepted model (i.e. high/low MARs over glacials/interglacials; Kohfeld & Harrison 2003), the average MARs for the MIS 5–MIS 3 period (~130–57 ka) show a contradictory trend: the MAR values during MIS 5 are only slightly lower than during MIS 4, while the MARs for MIS 3 are higher than those calculated for the MIS 2 period. It is possible that factors such as the vicinity of major rivers, specific geomorphology, more efficient dust trapping, and availability of potential secondary dust sources caused a near-continuous dust sedimentation during the last ~130 ka. However, it is more likely that the accuracy of the dust flux estimates is hampered by multiple factors such as low dating resolution, age distribution biases, systematic dating errors, and sampling position biases. Furthermore, the dynamic nature of the Danube River and its tributaries over the past five MI stages introduces additional uncertainty. Due to river channel migration influenced by geological and climatic shifts, it was not possible to determine the exact historical positions of these rivers and account for these shifts. Likewise, processes such as erosion, re-deposition, slope movement, and sheetwash likely affect MAR estimations. For example, erosional hiatuses can result in artificially low MARs by omitting certain layers, while re-deposition may inflate MARs by adding older material to specific layers. Given these complexities, the presented MAR values may not fully capture atmospheric dust flux variability for the entire MIS 5–MIS 1 period. It might be the case that the calculated MARs show average values over the dated time period at individual sites, rather than the actual regional atmospheric dust variability.

At this stage, it is debatable whether the presented MAR values for the last ~130 ka are a true representation of the atmospheric dust fluxes during this period. There are still too few high-resolution LPS chronologies available, which are necessary to construct reliable dust deposition models for the entire MIS 5–MIS 1 period. For a more reliable estimate of the dust flux trends in the Middle Danube Basin numerical independent chronologies with a higher dating resolution are needed, which would allow the detection of slight sedimentation rate changes, breaks, and erosion events. Until these are available, we recommend caution when interpreting atmospheric dust activity using currently available chronologies.

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Author contributions. – ZMP, SBM and RSM designed the study. ZMP collected the published data, performed the age-depth modelling and calculated the MARs. MGR calculated the distance of the investigated LPSs to the silt sources, determined the geographical position of the sites and modified the map. All authors contributed to the writing of the final manuscript.

Data availability. – The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- Aitken, M. J. 1998: *Introduction to Optical Dating: The Dating of Quaternary Sediments by the Use of Photon-Stimulated Luminescence*. 267 pp. Clarendon Press, Oxford.
- Aitken, M. J. 2014: *Science-Based Dating in Archaeology*. 4 pp. Routledge, London. <https://doi.org/10.4324/9781315836645>.
- Albani, S. & Mahowald, N. M. 2019: Paleodust insights into dust impacts on climate. *Journal of Climate* 32, 7897–7913. <https://doi.org/10.1175/JCLI-D-18-0742.1>.
- Albani, S., Mahowald, N. M., Winckler, G., Anderson, R. F., Bradtmiller, L. I., Delmonte, B., François, R., Goman, M., Heavens, N. G., Hesse, P. P., Hovan, S. A., Kang, S. G., Kohfeld, K. E., Lu, H., Maggi, V., Mason, J. A., Mayewski, P. A., McGee, D., Miao, X., Otto-Bliesner, B. J., Perry, A. T., Pourmand, A., Roberts, H. M., Rosenbloom, N., Stevens, T. & Sun, J. 2015: Twelve thousand years of dust: the Holocene global dust cycle constrained by natural archives. *Climate of the Past* 11, 869–903. <https://doi.org/10.5194/CP-11-869-2015>.
- Almond, P. C., Shanhan, F. L., Rieser, U. & Shulmeister, J. 2007: An OSL, radiocarbon and tephra isochron-based chronology for Birdlings Flat loess at Ahuriri Quarry, Banks Peninsula, Canterbury, New Zealand. *Quaternary Geochronology* 2, 4–8. <https://doi.org/10.1016/j.quageo.2006.06.002>.
- An, Z., Kukla, G. J., Porter, S. C. & Xiao, J. 1991: Magnetic susceptibility evidence of monsoon variation on the Loess Plateau of central China during the last 130,000 years. *Quaternary Research* 36, 29–36. [https://doi.org/10.1016/0033-5894\(91\)90015-W](https://doi.org/10.1016/0033-5894(91)90015-W).
- Antoine, P., Rousseau, D. D., Fuchs, M., Hatté, C., Gauthier, C., Marković, S. B., Jovanović, M., Gaudenyi, T., Moine, O. & Rossignol, J. 2009: High-resolution record of the last climatic cycle in the southern Carpathian Basin (Surduk, Vojvodina, Serbia). *Quaternary International* 198, 19–36. <https://doi.org/10.1016/J.QUAINT.2008.12.008>.
- Augustin, L. and 55 others. 2004: Eight glacial cycles from an Antarctic ice core. *Nature* 429, 623–628. <https://doi.org/10.1038/nature02599>.
- Avram, A., Constantin, D., Veres, D., Kelemen, S., Obrecht, I., Hambach, U., Marković, S. B. & Timar-Gabor, A. 2020: Testing polymineral post-IR IRSL and quartz SAR-OSL protocols on Middle to Late Pleistocene loess at Batajnica, Serbia. *Boreas* 49, 615–633. <https://doi.org/10.1111/bor.12442>.
- Balázs, A., Matenco, L., Magyar, I., Horváth, F. & Cloetingh, S. 2016: The link between tectonics and sedimentation in back-arc basins: new genetic constraints from the analysis of the Pannonian Basin. *Tectonics* 356, 1526–1559. <https://doi.org/10.1002/2015TC004109>.
- Basarin, B., Vandenbergh, D. A. G., Marković, S. B., Catto, N., Hambach, U., Vasiliniuc, S., Derese, C., Rončević, S., Vasiljević, D. A. & Rajić, L. 2011: The Belotinac section (Southern Serbia) at the southern limit of the European loess belt: initial results. *Quaternary International* 240, 128–138. <https://doi.org/10.1016/J.QUAINT.2011.02.022>.
- Blaauw, M. & Christen, J. A. 2011: Flexible paleoclimate age-depth models using an autoregressive gamma process. *Bayesian Analysis* 6, 457–474. <https://doi.org/10.1214/11-BA618>.

- Bokhorst, M. P., Beets, C. J., Marković, S. B., Gerasimenko, N. P., Matviishina, Z. N. & Frechen, M. 2009: Pedo-chemical climate proxies in late Pleistocene Serbian-Ukrainian loess sequences. *Quaternary International* 198, 113–123. <https://doi.org/10.1016/J.QUAINT.2008.09.003>.
- Bösken, J., Klasen, N., Zeeden, C., Obreht, I., Marković, S. B., Hambach, U. & Lehmkuhl, F. 2017: New luminescence-based geochronology framing the last two glacial cycles at the southern limit of European Pleistocene loess in Stalać (Serbia). *Geochronometria* 44, 150–161. <https://doi.org/10.1515/geochr-2015-0062>.
- Bösken, J., Obreht, I., Zeeden, C., Klasen, N., Hambach, U., Sümegi, P. & Lehmkuhl, F. 2019: High-resolution paleoclimatic proxy data from the MIS3/2 transition recorded in northeastern Hungarian loess. *Quaternary International* 502, 95–107. <https://doi.org/10.1016/J.QUAINT.2017.12.008>.
- Bridge, J. S. 2009: *Rivers and Floodplains: Forms, Processes, and Sedimentary Record*, 491 pp. Blackwell Publishing, Oxford.
- Buggle, B., Glaser, B., Zöller, L., Hambach, U., Marković, S., Glaser, I. & Gerasimenko, N. 2008: Geochemical characterization and origin of Southeastern and Eastern European loesses (Serbia, Romania, Ukraine). *Quaternary Science Reviews* 27, 1058–1075. <https://doi.org/10.1016/J.QUASCIREV.2008.01.018>.
- Buggle, B., Hambach, U., Kehl, M., Marković, S. B., Zöller, L. & Glaser, B. 2013: The progressive evolution of a continental climate in southeast-central European lowlands during the Middle Pleistocene recorded in loess palaeosol sequences. *Geology* 41, 771–774. <https://doi.org/10.1130/G34198.1>.
- Buylaert, J.-P., Jain, M., Murray, A. S., Thomsen, K. J., Thiel, C. & Sohbati, R. 2012: A robust feldspar luminescence dating method for Middle and Late Pleistocene sediments. *Boreas* 41, 435–451. <https://doi.org/10.1111/j.1502-3885.2012.00248.x>.
- Buylaert, J.-P., Thiel, C., Murray, A. S., Vandenberghe, D. A. G., Yi, S. & Lu, H. 2011: IRSL and post-IR IRSL residual doses recorded in modern dust samples from the Chinese loess plateau. *Geochronometria* 38, 432. <https://doi.org/10.2478/s13386-011-0047-0>.
- Buylaert, J. P., Yeo, E. Y., Thiel, C., Yi, S., Stevens, T., Thompson, W., Frechen, M., Murray, A. & Lu, H. 2015: A detailed post-IR IRSL chronology for the last interglacial soil at the Jingbian loess site (northern China). *Quaternary Geochronology* 30, 194–199. <https://doi.org/10.1016/j.quageo.2015.02.022>.
- Claquin, T., Roelandt, C., Kohfeld, K. E., Harrison, S. P., Tegen, I., Prentice, I. C., Balkanski, Y., Bergametti, G., Hansson, M., Mahowald, N., Rodhe, H. & Schulz, M. 2003: Radiative forcing of climate by ice-age atmospheric dust. *Climate Dynamics* 20, 193–202. <https://doi.org/10.1007/S00382-002-0269-1>.
- Derbyshire, E. 2003: Loess, and the Dust Indicators and Records of Terrestrial and Marine Palaeoenvironments (DIRTMAP) database. *Quaternary Science Reviews* 22, 1813–1819. [https://doi.org/10.1016/S0277-3791\(03\)00209-9](https://doi.org/10.1016/S0277-3791(03)00209-9).
- Ducić, V. & Radovanović, M. 2005: *Climate of Serbia*. 91 pp. Institute for Textbooks and Teaching Aids, Belgrade.
- Fenn, K., Millar, I. L., Durcan, J. A., Thomas, D. S. G., Banak, A., Marković, S. B., Veres, D. & Stevens, T. 2022: The provenance of Danubian loess. *Earth-Science Reviews* 226, 103920. <https://doi.org/10.1016/j.earscirev.2022.103920>.
- Fischer, P., Jöris, O., Fitzsimmons, K. E., Vinnepand, M., Prud'homme, C., Schulte, P., Hatté, C., Hambach, U., Lindauer, S., Zeeden, C., Peric, Z., Lehmkuhl, F., Wunderlich, T., Wilken, D., Schirmer, W. & Vött, A. 2021: Millennial-scale terrestrial ecosystem responses to Upper Pleistocene climatic changes: 4D-reconstruction of the Schwalbenberg Loess-Palaeosol-Sequence (Middle Rhine Valley, Germany). *Catena* 196, 104913. <https://doi.org/10.1016/J.CATENA.2020.104913>.
- Frechen, M., Oches, E. A. & Kohfeld, K. E. 2003: Loess in Europe—mass accumulation rates during the Last Glacial Period. *Quaternary Science Reviews* 22, 1835–1857. [https://doi.org/10.1016/S0277-3791\(03\)00183-5](https://doi.org/10.1016/S0277-3791(03)00183-5).
- Fuchs, M., Rousseau, D. D., Antoine, P., Hatté, C., Gauthier, C., Marković, S. & Zoeller, L. 2008: Chronology of the Last Climatic Cycle (Upper Pleistocene) of the Surduk loess sequence, Vojvodina, Serbia. *Boreas* 37, 66–73. <https://doi.org/10.1111/J.1502-3885.2007.00012.X>.
- Galović, L., Frechen, M., Halamić, J., Durn, G. & Romić, M. 2009: Loess chronostratigraphy in Eastern Croatia—a luminescence dating approach. *Quaternary International* 198, 85–97. <https://doi.org/10.1016/J.QUAINT.2008.02.004>.
- Haas, J. 2015: Geological and tectonic background. In Lóczy, D. (ed.): *Landscapes and Landforms of Hungary*, 7–17. Springer International Publishing. https://doi.org/10.1007/978-3-319-08997-3_2.
- Hatté, C., Gauthier, C., Rousseau, D. D., Antoine, P., Fuchs, M., Lagroix, F., Marković, S. B., Moine, O. & Sima, A. 2013: Excursions to C4 vegetation recorded in the upper pleistocene loess of surduk (Northern Serbia): An organic isotope geochemistry study. *Climate of the Past* 9, 1001–1014. <https://doi.org/10.5194/CP-9-1001-2013>.
- Huntley, D. J. & Lamothe, M. 2001: Ubiquity of anomalous fading in K-feldspars and the measurement and correction for it in optical dating. *Canadian Journal of Earth Sciences* 38, 1093–1106. <https://doi.org/10.1139/e01-013>.
- Jacoby, W. G. 2000: Loess: a nonparametric, graphical tool for depicting relationships between variables. *Electoral Studies* 19, 577–613. [https://doi.org/10.1016/S0261-3794\(99\)00028-1](https://doi.org/10.1016/S0261-3794(99)00028-1).
- Jovanovic, M., Markovic, S. B., Gaudenyi, T., Oches, E. A., Hambach, U. & Zoeller, L. 2006: 'Warm' glacial climate during loess deposition recorded at exposures of the Pozarevac brickyard, NE Serbia. *Geophysical Research Abstracts* 8, EGU06-A-10599.
- Juhász, G., Pogácsás, G., Magyar, I. & Vakarc, G. 2007: Tectonic versus climatic control on the evolution of fluvio-deltaic systems in a lake basin, Eastern Pannonian Basin. *Sedimentary Geology* 202, 72–95. <https://doi.org/10.1016/j.sedgeo.2007.05.001>.
- Kang, S., Roberts, H. M., Wang, X., An, Z. & Wang, M. 2015: Mass accumulation rate changes in Chinese loess during MIS 2, and asynchrony with records from Greenland ice cores and North Pacific Ocean sediments during the Last Glacial Maximum. *Aeolian Research* 19, 251–258. <https://doi.org/10.1016/J.AEOLIA.2015.05.005>.
- Kapp, P., Pullen, A., Pelletier, J. D., Russell, J., Goodman, P. & Cai, F. 2015: From dust to dust: quaternary wind erosion of the Mu Us Desert and Loess Plateau, China. *Geology* 43, 835–838. <https://doi.org/10.1130/G36724.1>.
- Kemp, R. A. 2001: Pedogenic modification of loess: significance for palaeoclimatic reconstructions. *Earth-Science Reviews* 54, 145–156. [https://doi.org/10.1016/S0012-8252\(01\)00045-9](https://doi.org/10.1016/S0012-8252(01)00045-9).
- Kohfeld, K. E. 2000: 3. Late Quaternary Aeolian Deposition on the Chinese Loess Plateau. *Workshop on the Role of Mineral Aerosols in Quaternary Climate Cycles—Models and Data*, 8–11 October 2000. Available at: <https://d-nb.info/117767016X/34#page=23>.
- Kohfeld, K. E. & Harrison, S. P. 2001: DIRTMAP: the geological record of dust. *Earth-Science Reviews* 54, 81–114. [https://doi.org/10.1016/S0012-8252\(01\)00042-3](https://doi.org/10.1016/S0012-8252(01)00042-3).
- Kohfeld, K. E. & Harrison, S. P. 2003: Glacial-interglacial changes in dust deposition on the Chinese Loess Plateau. *Quaternary Science Reviews* 22, 1859–1878. [https://doi.org/10.1016/S0277-3791\(03\)00166-5](https://doi.org/10.1016/S0277-3791(03)00166-5).
- Kukla, G. J. 1977: Pleistocene land-sea correlations I. Europe. *Earth Science Reviews* 13, 307–374. [https://doi.org/10.1016/0012-8252\(77\)90125-8](https://doi.org/10.1016/0012-8252(77)90125-8).
- Lambert, F., Delmonte, B., Petit, J. R., Bigler, M., Kaufmann, P. R., Hutterli, M. A., Stocker, T. F., Ruth, U., Steffensen, J. P. & Maggi, V. 2008: Dust-climate couplings over the past 800,000 years from the EPICA Dome C ice core. *Nature* 452, 7187. <https://doi.org/10.1038/nature06763>.
- Lamothe, M., Auclair, M., Hamzaoui, C. & Huot, S. 2003: Towards a prediction of long-term anomalous fading of feldspar IRSL. *Radiation Measurements* 37, 493–498. [https://doi.org/10.1016/S1350-4487\(03\)00016-7](https://doi.org/10.1016/S1350-4487(03)00016-7).
- Lehmkuhl, F., Bösken, J., Hošek, J., Sprafke, T., Marković, S. B., Obreht, I., Hambach, U., Sümegi, P., Thiemann, A., Steffens, S., Lindner, H., Veres, D. & Zeeden, C. 2018: Loess distribution and related Quaternary sediments in the Carpathian Basin. *Journal of Maps* 14, 661–670. <https://doi.org/10.1080/17445647.2018.1526720>.
- Lehmkuhl, F., Nett, J. J., Pötter, S., Schulte, P., Sprafke, T., Jary, Z., Antoine, P., Wacha, L., Wolf, D., Zerboni, A., Hošek, J., Marković, S. B., Obreht, I., Sümegi, P., Veres, D., Zeeden, C., Boemke, B., Schaubert, V., Viehweger, J. & Hambach, U. 2021: Loess landscapes

- of Europe – mapping, geomorphology, and zonal differentiation. *Earth-Science Reviews* 215, 103496. <https://doi.org/10.1016/j.earscirev.2020.103496>.
- Licht, A., Pullen, A., Kapp, P., Abell, J. & Giesler, N. 2016: Eolian cannibalism: reworked loess and fluvial sediment as the main sources of the Chinese Loess Plateau. *GSA Bulletin* 128, 944–956. <https://doi.org/10.1130/B31375.1>.
- Lisiecki, L. E. & Raymo, M. E. 2005: A Pliocene-Pleistocene stack of 57 globally distributed benthic $\delta^{18}O$ records. *Paleoceanography* 20, PA1003. <https://doi.org/10.1029/2004PA001071>.
- Liu, T. & Ding, Z. 1998: Chinese loess and the paleomonsoon. *Annual Review of Earth and Planetary Sciences* 26, 111–145. <https://doi.org/10.1146/ANNUREV.EARTH.26.1.111>.
- Lougheed, B. C. & Obrochta, S. P. 2019: A rapid, deterministic age-depth modeling routine for geological sequences with inherent depth uncertainty. *Paleoceanography and Paleoclimatology* 34, 122–133. <https://doi.org/10.1029/2018PA003457>.
- Marković, S. B., Bokhorst, M. P., Vandenberghe, J., McCoy, W. D., Oches, E. A., Hambach, U., Gaudenyi, T., Jovanović, M., Zöller, L., Stevens, T. & Machallet, B. 2008: Late Pleistocene loess-paleosol sequences in the Vojvodina region, north Serbia. *Journal of Quaternary Science* 23, 73–84. <https://doi.org/10.1002/JQS.1124>.
- Marković, S. B., Hambach, U., Catto, N., Jovanović, M., Buggle, B., Machallet, B., Zöller, L., Glaser, B. & Frechen, M. 2009: Middle and late Pleistocene loess sequences at Batajnica, Vojvodina, Serbia. *Quaternary International* 198, 255–266.
- Marković, S. B., Oches, E. A., McCoy, W. D., Frechen, M. & Gaudenyi, T. 2007: Malacological and sedimentological evidence for “warm” glacial climate from the Irig loess sequence, Vojvodina, Serbia. *Geochemistry, Geophysics, Geosystems* 8, Q09008. <https://doi.org/10.1029/2006GC001565>.
- Marković, S. B., Oches, E. A., Perić, Z. M., Gaudenyi, T., Jovanović, M., Sipos, G., Thiel, C., Buylaert, J.-P., Savić, S., McCoy, W. D., Radaković, M. G., Marković, R. S. & Gavrilov, M. B. 2021: The Požarevac loess-paleosol sequence: a record of increased aridity in the south-eastern margin of the Carpathian Basin during the last 350 ka. *Journal of Quaternary Science* 8, 1436–1447. <https://doi.org/10.1002/jqs.3327>.
- Marković, S. B., Oches, E., Sümegi, P., Jovanović, M. & Gaudenyi, T. 2006: An introduction to the Middle and Upper Pleistocene loess–paleosol sequence at Ruma brickyard, Vojvodina, Serbia. *Quaternary International* 149, 80–86. <https://doi.org/10.1016/j.quaint.2005.11.020>.
- Marković, S. B., Stevens, T., Kukla, G. J., Hambach, U., Fitzsimmons, K. E., Gibbard, P., Buggle, B., Zech, M., Guo, Z., Hao, Q., Wu, H., O'Hara Dhand, K., Smalley, I. J., Újvári, G., Sümegi, P., Timar-Gabor, A., Veres, D., Sirocko, F., Vasiljević, D. A., Jary, Z. & Svirčev, Z. 2015: Danube loess stratigraphy—towards a pan-European loess stratigraphic model. *Earth-Science Reviews* 148, 228–258. <https://doi.org/10.1016/j.earscirev.2015.06.005>.
- Marković, S. B., Sümegi, P., Stevens, T., Schaetzl, R. J., Obrecht, I., Chu, W., Buggle, B., Zech, M., Zech, R., Zeeden, C., Gavrilov, M. B., Perić, Z., Svirčev, Z. & Lehmkuhl, F. 2018: The Crvenka loess-paleosol sequence: a record of continuous grassland domination in the southern Carpathian Basin during the Late Pleistocene. *Palaeogeography, Palaeoclimatology, Palaeoecology* 509, 33–46. <https://doi.org/10.1016/j.palaeo.2018.03.019>.
- Marković, S. B., Timar-Gabor, A., Stevens, T., Hambach, U., Popov, D., Tomić, N., Obrecht, I., Jovanović, M., Lehmkuhl, F., Kels, H., Marković, R. & Gavrilov, M. B. 2014: Environmental dynamics and luminescence chronology from the Orlovat loess–paleosol sequence (Vojvodina, northern Serbia). *Journal of Quaternary Science* 29, 189–199. <https://doi.org/10.1002/jqs.2693>.
- Marković, S. B., Vandenberghe, J., Perić, Z. M., Filyó, D., Bartyik, T., Radaković, M. G., Hao, Q., Marković, R. S., Lukić, T., Tomić, N., Gavrilov, M. B., Antić, A., Cvijanović, I. & Sipos, G. 2023: Local differentiation in the loess deposition as a function of dust source: key study novo Orahovo loess Paleosol sequence (Vojvodina, Serbia). *Quaternary* 6, 1. <https://doi.org/10.3390/quat6010023>.
- McGee, D., Broecker, W. S. & Winckler, G. 2010: Gustiness: the driver of glacial dustiness? *Quaternary Science Reviews* 29, 2340–2350. <https://doi.org/10.1016/j.quascirev.2010.06.009>.
- Mellars, P. 2006: A new radiocarbon revolution and the dispersal of modern humans in Eurasia. *Nature* 439, 931–935. <https://doi.org/10.1038/nature04521>.
- Muhs, D. R. 2013: Loess and its geomorphic, stratigraphic and paleoclimatic significance in the Quaternary. *Aeolian Geomorphology* 11, 149–183. <https://doi.org/10.1016/B978-0-12-374739-6.00302-X>.
- Murray, A. S., Thomsen, K. J., Masuda, N., Buylaert, J. P. & Jain, M. 2012: Identifying well-bleached quartz using the different bleaching rates of quartz and feldspar luminescence signals. *Radiation Measurements* 47, 688–695. <https://doi.org/10.1016/j.radmeas.2012.05.006>.
- Nemeth, K. & Martin, U. 1999: Large hydrovolcanic field in the Pannonian Basin: general characteristics of the Bakony-Balaton Highland Volcanic Field, Hungary. *Acta Vulcanologica* 11, 271–282.
- Novothy, Á., Barta, G., Végh, T., Bradák, B., Surányi, G. & Horváth, E. 2020: Correlation of drilling cores and the Paks brickyard key section at the area of Paks, Hungary. *Quaternary International* 552, 50–61. <https://doi.org/10.1016/j.quaint.2019.09.012>.
- Novothy, Á., Frechen, M., Horváth, E., Wacha, L. & Rolf, C. 2011: Investigating the penultimate and last glacial cycles of the Süttő loess section (Hungary) using luminescence dating, high-resolution grain size, and magnetic susceptibility data. *Quaternary International* 234, 75–85. <https://doi.org/10.1016/j.quaint.2010.08.002>.
- Open Maps for Europe/Eurogeographics. n.d. Available at: <https://www.mapsforeurope.org/datasets/euro-dem> (accessed 18.05.2024).
- Pécsi, M. 1990: Loess is not just the accumulation of dust. *Quaternary International* 7–8, 1–21. [https://doi.org/10.1016/1040-6182\(90\)90034-2](https://doi.org/10.1016/1040-6182(90)90034-2).
- Pécsi, M. & Richter, G. 1996: Löss. Herkunft—Gliederung—Landschaften. *Zeitschrift für Geomorphologie. Supplementband* 98, 391. <http://pascal-francis.inist.fr/vibad/index.php?action=getRecordDetail&idt=6313664>.
- Perić, Z., Lagerbäck Adolphi, E., Stevens, T., Újvári, G., Zeeden, C., Buylaert, J. P., Marković, S. B., Hambach, U., Fischer, P., Schmidt, C., Schulte, P., Huayu, L., Shuangwen, Y., Lehmkuhl, F., Obrecht, I., Veres, D., Thiel, C., Frechen, M., Jain, M., Vött, A., Zöller, L. & Gavrilov, M. B. 2019: Quartz OSL dating of late quaternary Chinese and Serbian loess: a cross Eurasian comparison of dust mass accumulation rates. *Quaternary International* 502, 30–44. <https://doi.org/10.1016/J.QUANT.2018.01.010>.
- Perić, Z. M., Marković, S. B., Avram, A., Timar-Gabor, A., Zeeden, C., Nett, J. J., Fischer, P., Fitzsimmons, K. E. & Gavrilov, M. B. 2022a: Initial quartz OSL and dust mass accumulation rate investigation of the Kisiljevo loess sequence in north-eastern Serbia. *Quaternary International* 620, 13–23. <https://doi.org/10.1016/J.QUANT.2020.10.040>.
- Perić, Z. M., Marković, S. B., Filyó, D., Thiel, C., Murray, A. S., Gavrilov, M. B., Nett, J. J. & Sipos, G. 2021: Quartz OSL and polymineral post IR–IRSL dating of the Požarevac loess–paleosol sequence in north-eastern Serbia. *Quaternary Geochronology* 66, 101216. <https://doi.org/10.1016/J.QUAGEO.2021.101216>.
- Perić, Z. M., Marković, S. B., Sipos, G., Gavrilov, M. B., Thiel, C., Zeeden, C. & Murray, A. S. 2020: A post-IR IRSL chronology and dust mass accumulation rates of the Nosak loess-paleosol sequence in northeastern Serbia. *Boreas* 49, 841–857. <https://doi.org/10.1111/bor.12459>.
- Perić, Z. M., Stevens, T., Obrecht, I., Hambach, U., Lehmkuhl, F. & Marković, S. B. 2022b: Detailed luminescence dating of dust mass accumulation rates over the last two glacial-interglacial cycles from the Irig loess-paleosol sequence, Carpathian Basin. *Global and Planetary Change* 215, 103895. <https://doi.org/10.1016/j.gloplacha.2022.103895>.
- Perić, Z. M., Stevens, T., Obrecht, I. & Marković, S. B. 2022c: Dataset of dust mass accumulation rates for the loess-paleosol sequences from the Carpathian Basin. *Data in Brief* 44, 108555. <https://doi.org/10.1016/j.dib.2022.108555>.
- Perić, Z. M., Ryan, C., Alexanderson, H. & Marković, S. B. 2024: Revised OSL chronology of the Kisiljevo loess-paleosol sequence: New insight into the dust flux in the eastern Carpathian Basin during

- MIS 3 - MIS1. *Quaternary International*. <https://doi.org/10.1016/j.quaint.2024.06.006>.
- Porter, S. C. & An, Z. 1995: Correlation between climate events in the North Atlantic and China during the last glaciation. *Nature* 375, 6529. <https://doi.org/10.1038/375305a0>.
- Prescott, J. R. & Robertson, G. B. 1997: Sediment dating by luminescence: a review. *Radiation Measurements* 27, 893–922. [https://doi.org/10.1016/S1350-4487\(97\)00204-7](https://doi.org/10.1016/S1350-4487(97)00204-7).
- Pye, K. & Tsoar, H. 2009: *Aeolian Sand and Sand Dunes*. 251 pp. Springer-Verlag, Heidelberg. <https://doi.org/10.1007/978-3-540-85910-9>.
- Radaković, M. G., Oches, E. A., Hughes, P. D., Marković, R. S., Hao, Q., Perić, Z. M., Gavrilović, B., Ludwig, P., Lukić, T., Gavrilov, M. B. & Marković, S. B. 2023: Reconstructed Malacothermometer July Paleotemperatures from the Last Nine Glacials over the South-Eastern Carpathian Basin (Serbia). *Atmosphere* 14, 5. <https://doi.org/10.3390/atmos14050791>.
- Roberts, H. M. 2012: Testing Post-IR IRSL protocols for minimising fading in feldspars, using Alaskan loess with independent chronological control. *Radiation Measurements* 47, 716–724. <https://doi.org/10.1016/J.RADMEAS.2012.03.022>.
- Rousseau, D.-D., Derbyshire, E., Antoine, P. & Hatté, C. 2018: European loess records. *Earth System and Environment Science*, 1–17. <https://doi.org/10.7916/D8XH17TM>.
- Ruth, U., Wagenbach, D., Steffensen, J. P. & Bigler, M. 2003: Continuous record of microparticle concentration and size distribution in the central Greenland NGRIP ice core during the last glacial period. *Journal of Geophysical Research: Atmospheres* 108, 4098. <https://doi.org/10.1029/2002JD002376>.
- Schatz, A.-K., Buylaert, J.-P., Murray, A., Stevens, T. & Scholten, T. 2012: Establishing a luminescence chronology for a palaeosol-loess profile at Tokaj (Hungary): a comparison of quartz OSL and polymineral IRSL signals. *Quaternary Geochronology* 10, 68–74. <https://doi.org/10.1016/j.quageo.2012.02.018>.
- Schmid, S. M., Bernoulli, D., Fügenschuh, B., Matenco, L., Schefer, S., Schuster, R., Tischler, M. & Ustaszewski, K. 2008: The Alpine-Carpathian-Dinaridic orogenic system: correlation and evolution of tectonic units. *Swiss Journal of Geosciences* 101, 139–183. <https://doi.org/10.1007/s00015-008-1247-3>.
- Schmidt, E. D., Machalett, B., Marković, S. B., Tsukamoto, S. & Frechen, M. 2010: Luminescence chronology of the upper part of the Stari Slankamen loess sequence (Vojvodina, Serbia). *Quaternary Geochronology* 5, 137–142. <https://doi.org/10.1016/J.QUAGEO.2009.09.006>.
- Scelate, J. G., Royden, L., Horvath, F., Burchfiel, B. C., Semken, S. & Stegena, L. 1980: The formation of the intra-Carpathian basins as determined from subsidence data. *51*, 139–162. [https://doi.org/10.1016/0012-821X\(80\)90262-9](https://doi.org/10.1016/0012-821X(80)90262-9).
- Selsing, L. & Mejdahl, V. 1994: Aeolian stratigraphy and thermoluminescence dating of sediments of late Holocene age from Sola, southwest Norway. *Boreas* 23, 92–104. <https://doi.org/10.1111/j.1502-3885.1994.tb00589.x>.
- Sier, M. J., Peeters, J., Dekkers, M. J., Parés, J. M., Chang, L., Busschers, F. S., Cohen, K. M., Wallinga, J., Bunnik, F. P. M. & Roebroeks, W. 2015: The Blake Event recorded near the Eemian type locality – a diachronic onset of the Eemian in Europe. *Quaternary Geochronology* 28, 12–28. <https://doi.org/10.1016/j.quageo.2015.03.003>.
- Smalley, I. J., O'Hara-Dhand, K., Wint, J., Machalett, B., Jary, Z. & Jefferson, I. 2009: Rivers and loess: the significance of long river transportation in the complex event-sequence approach to loess deposit formation. *Quaternary International* 198, 7–18. <https://doi.org/10.1016/j.quaint.2008.06.009>.
- Smith, B. W., Rhodes, E. J., Stokes, S., Spooner, N. A. & Aitken, M. J. 1990: Optical dating of sediments: initial quartz results from Oxford. *Archaeometry* 32, 19–31. <https://doi.org/10.1111/j.1475-4754.1990.tb01078.x>.
- Song, Y., Lai, Z., Li, Y., Chen, T. & Wang, Y. 2015: Comparison between luminescence and radiocarbon dating of late Quaternary loess from the Ili Basin in Central Asia. *Quaternary Geochronology* 30, 405–410. <https://doi.org/10.1016/j.quageo.2015.01.012>.
- Stevens, T., Buylaert, J. P., Lu, H., Thiel, C., Murray, A., Frechen, M., Yi, S. & Zeng, L. 2016: Mass accumulation rate and monsoon records from Xifeng, Chinese Loess Plateau, based on a luminescence age model. *Journal of Quaternary Science* 31, 391–405. <https://doi.org/10.1002/JQS.2848>.
- Stevens, T., Buylaert, J.-P., Thiel, C., Újvári, G., Yi, S., Murray, A. S., Frechen, M. & Lu, H. 2018: Ice-volume-forced erosion of the Chinese Loess Plateau global Quaternary stratotype site. *Nature Communications* 9, 983. <https://doi.org/10.1038/s41467-018-03329-2>.
- Stevens, T., Marković, S. B., Zech, M., Hambach, U. & Sümege, P. 2011: Dust deposition and climate in the Carpathian Basin over an independently dated last glacial-interglacial cycle. *Quaternary Science Reviews* 30, 662–681. <https://doi.org/10.1016/j.quascirev.2010.12.011>.
- Stokes, S. & Gaylord, D. R. 1993: Optical dating of Holocene Dune Sands in the Ferris Dune Field, Wyoming. *Quaternary Research* 39, 274–281. <https://doi.org/10.1006/qres.1993.1034>.
- Street, M., Baales, M., Czesla, E., Hartz, S., Heinen, M., Jöris, O., Koch, I., Pasda, C., Terberger, T. & Vollbrecht, J. 2001: Final Paleolithic and Mesolithic research in reunified Germany. *Journal of World Prehistory* 15, 365–453. <https://doi.org/10.1023/A:1014332527763>.
- Sümege, P., Gulyás, S., Molnár, D., Szilágyi, G., Sümege, B. P., Töröcsik, T. & Molnár, M. 2020: 14C dated chronology of the thickest and best resolved loess/paleosol record of the LGM from SE Hungary based on comparing precision and accuracy of age-depth models. *Radiocarbon* 62 (2), 403–417. <https://doi.org/10.1017/RDC.2019.154>.
- Sümege, P. & Krolopp, E. 2002: Quaternary malacological analyses for modeling of the Upper Weichselian palaeoenvironmental changes in the Carpathian Basin. *Quaternary International* 91, 53–63. [https://doi.org/10.1016/S1040-6182\(01\)00102-1](https://doi.org/10.1016/S1040-6182(01)00102-1).
- Sümege, P., Magyari, E., Dániel, P., Molnár, M. & Töröcsik, T. 2013: Responses of terrestrial ecosystems to Dansgaard-Oeschger cycles and Heinrich events: A 28,000-year record of environmental changes from SE Hungary. *Quaternary International* 293, 34–50. <https://doi.org/10.1016/j.quaint.2012.07.032>.
- Sümege, P., Marković, S. B., Molnár, D., Sávai, S., Náfrádi, K., Szelepcsényi, Z. & Novák, Z. 2016: Crvenka loess-paleosol sequence revisited: local and regional quaternary biogeographical inferences of the southern Carpathian Basin. *Open Geosciences* 8, 390–404. <https://doi.org/10.1515/geo-2016-0031>.
- Sümege, P., Molnár, D., Gulyás, S., Náfrádi, K., Sümege, B. P., Töröcsik, T., Persaits, G., Molnár, M., Vandenberghe, J. & Zhou, L. 2019: High-resolution proxy record of the environmental response to climatic variations during transition MIS3/MIS2 and MIS2 in Central Europe: The loess-paleosol sequence of Katymár brickyard (Hungary). *Quaternary International* 504, 40–55. <https://doi.org/10.1016/j.quaint.2018.03.030>.
- Sümege, P., Molnár, D., Gulyás, S., Stevens, T., Makó, L., Cseh, P., Molnár, M., Fitzsimmons, K., Nett, J. J., Hlavatskyi, D. & Lehmkuhl, F. 2022: Comparison of high-resolution 14C and luminescence-based chronologies of the MIS 2 Madaras loess/paleosol sequence, Hungary: implications for chronological studies. *Quaternary* 5, 4. <https://doi.org/10.3390/quat5040047>.
- Sümege, P., Molnár, M., Svingor, É., Szántó, Z., Hum, L. & Gulyás, S. 2007: Results of radiocarbon analysis of upper Weichselian loess sequences from Hungary. *Radiocarbon* 49, 1023–1030. <https://doi.org/10.1017/S0033822200042880>.
- Sümege, P., Náfrádi, K., Molnár, D. & Sávai, S. 2015: Results of paleoecological studies in the loess region of Szeged-Óthalom (SE Hungary). *Quaternary International* 372, 66–78. <https://doi.org/10.1016/j.quaint.2014.09.003>.
- Svirčev, Z., Dulić, T., Obreht, I., Codd, G. A., Lehmkuhl, F., Marković, S. B., Hambach, U. & Meriluoto, J. 2019: Cyanobacteria and loess—An underestimated interaction. *Plant and Soil* 439, 293–308. <https://doi.org/10.1007/s11104-019-04048-3>.
- Svirčev, Z., Marković, S. B., Stevens, T., Codd, G. A., Smalley, I., Simeunović, J., Obreht, I., Dulić, T., Pantelić, D. & Hambach, U. 2013: Importance of biological loess crusts for loess formation in semi-arid environments. *Quaternary International* 296, 206–215. <https://doi.org/10.1016/j.quaint.2012.10.048>.
- Thiel, C., Buylaert, J. P., Murray, A., Terhorst, B., Hofer, I., Tsukamoto, S. & Frechen, M. 2011: Luminescence dating of the Stratzing loess

- profile (Austria)—testing the potential of an elevated temperature post-IR IRSI protocol. *Quaternary International* 234, 23–31. <https://doi.org/10.1016/j.quaint.2010.05.018>.
- Timar-Gabor, A., Constantin, D., Marković, S. B. & Jain, M. 2015: Extending the area of investigation of fine versus coarse quartz optical ages from the Lower Danube to the Carpathian Basin. *Quaternary International* 388, 168–176. <https://doi.org/10.1016/j.quaint.2014.09.065>.
- Újvári, G., Kovács, J., Varga, G., Raucsik, B. & Marković, S. B. 2010: Dust flux estimates for the Last Glacial Period in East Central Europe based on terrestrial records of loess deposits: a review. *Quaternary Science Reviews* 29, 3157–3166. <https://doi.org/10.1016/j.quascirev.2010.07.005>.
- Újvári, G., Molnár, M., Novothny, Á., Páll-Gergely, B., Kovács, J. & Várhegyi, A. 2014: AMS 14C and OSL/IRSL dating of the Dunaszekcső loess sequence (Hungary): chronology for 20 to 150 ka and implications for establishing reliable age–depth models for the last 40 ka. *Quaternary Science Reviews* 106, 140–154. <https://doi.org/10.1016/j.quascirev.2014.06.009>.
- Újvári, G., Stevens, T., Molnár, M., Demény, A., Lambert, F., Varga, G., Timothy Jull, A. J., Páll-Gergely, B., Buylaert, J. P. & Kovács, J. 2017: Coupled European and Greenland last glacial dust activity driven by North Atlantic climate. *Proceedings of the National Academy of Sciences of the United States of America* 114, E10632–E10638. <https://doi.org/10.1073/PNAS.1712651114>.
- Újvári, G., Varga, A. & Balogh-Brunstad, Z. 2008: Origin, weathering, and geochemical composition of loess in southwestern Hungary. *Quaternary Research* 69, 421–437. <https://doi.org/10.1016/j.yqres.2008.02.001>.
- Újvári, G., Varga, A., Ramos, F. C., Kovács, J., Németh, T. & Stevens, T. 2012: Evaluating the use of clay mineralogy, Sr–Nd isotopes and zircon U–Pb ages in tracking dust provenance: An example from loess of the Carpathian Basin. *Chemical Geology* 304–305, 83–96. <https://doi.org/10.1016/j.chemgeo.2012.02.007>.
- Vandenbergh, J., Marković, S. B., Jovanović, M. & Hambach, U. 2014: Site-specific variability of loess and palaeosols (Ruma, Vojvodina, northern Serbia). *Quaternary International* 334–335, 86–93. <https://doi.org/10.1016/j.quaint.2013.10.036>.
- van Vliet-Lanoë, B., Magyari, A. & Meilliez, F. 2004: Distinguishing between tectonic and periglacial deformations of quaternary continental deposits in Europe. *Global and Planetary Change* 43, 103–127. <https://doi.org/10.1016/j.gloplacha.2004.03.003>.
- Wacha, L. & Frechen, M. 2011: The geochronology of the “Gorjanović loess section” in Vukovar, Croatia. *Quaternary International* 240, 87–99. <https://doi.org/10.1016/j.quaint.2011.04.010>.
- Zech, R., Zech, M., Marković, S., Hambach, U. & Huang, Y. 2013: Humid glacials, arid interglacials? Critical thoughts on pedogenesis and paleoclimate based on multi-proxy analyses of the loess–paleosol sequence Crvenka, Northern Serbia. *Palaeogeography, Palaeoclimatology, Palaeoecology* 387, 165–175. <https://doi.org/10.1016/j.palaeo.2013.07.023>.
- Zeeden, C., Hambach, U., Obrecht, I., Hao, Q., Abels, H. A., Veres, D., Lehmkuhl, F., Gavrillo, M. B. & Marković, S. B. 2018: Patterns and timing of loess–paleosol transitions in Eurasia: constraints for paleoclimate studies. *Global and Planetary Change* 162, 1–7. <https://doi.org/10.1016/j.gloplacha.2017.12.021>.
- Zhao, H., Chen, F.-H., Li, S.-H., Wintle, A. G., Fan, Y.-X. & Xia, D.-S. 2007: A record of Holocene climate change in the Guanzhong Basin, China, based on optical dating of a loess–paleosol sequence. *The Holocene* 17, 1015–1022. <https://doi.org/10.1177/0959683607080530>.

Supporting Information

Additional Supporting Information to this article is available at <http://www.boreas.dk>.

Fig. S1. r.bacon (Blaauw & Christen 2011) generated age–depth Bayesian models for the LPSs in Hungary: (a) Bodrogkeresztúr I; (b) Csörgökút I; (c) Csörgökút

II; (d) Debrecen brickyard; (e) Dunaszekcső; (f) Hortobágy, Szálkahalom; (g) Kardoskút, Fehér-lake; (h) Katymar; (i) Lakiteleki brickyard; (j) Madaras; (k) Paks I core; (l) Paks II core; (m) Süttő; (n) Szeged-Óthalom; (o) Tiszaalpári high-bluff; (p) Tokaj Patkó-quarry; (q) Tokaj Kereszt Hill; (r) Udvari borehole. The results are based on millions of Monte Carlo iterations yielding a mean confidence of 95%. Top panel: time series of the log-posterior for the subsampled MCMC (left), prior (green) and posterior (grey) for accumulation rates.

Fig. S2. r.bacon generated age–depth models for the LPSs in Serbia: (a) Batajnica; (b) Belotinac; (c) Crvenka; (d) Irig; (e) Kisiljevo; (f) Nosak; (g) Novo Orahovo; (h) Orlovat; (i) Požarevac; (j) Stalać; (k) Stari Slankamen; (l) Surduk; (m) Veliki Surduk. The results are based on millions of Monte Carlo iterations yielding a mean confidence of 95%. Top panel: time series of the log-posterior for the subsampled MCMC (left), prior (green) and posterior (grey) for accumulation rates.

Fig. S3. r.bacon generated age–depth models for the LPSs in Croatia: (a) Erdut; (b) Vukovar; (c) Zmajevac. The results are based on millions of Monte Carlo iterations yielding a mean confidence of 95%. Top panel: time series of the log-posterior for the subsampled MCMC (left), prior (green) and posterior (grey) for accumulation rates.

Fig. S4. Dust mass accumulation rate (MAR) as a function of age for the Hungarian LPSs. The MARs were attained using the r.bacon model. The curves show the MARs according to the mean ages within 95% confidence range. The confidence range is based on quantiles, 2.5 and 97.5%, and is an indication of how well the model fits the dates (the percentage of dates that overlaps with the age–depth model at the set probability – default 95%) (for details see Blaauw & Christen 2011). For a more realistic representation of the MARs, the results were smoothed using the loess smoothing technique, with tricube weighting and polynomial regression using the SigmaPlot software version 11.0 (sampling proportion = 0.100; polynomial degree = 1). The grey lines represent the raw, unsmoothed MARs while the black lines show the smoothed MAR values.

Fig. S5. Dust mass accumulation rate (MAR) as a function of age for the LPSs situated in Serbia. The MARs were attained using the r.bacon model. The curves show the MARs according to the mean ages within 95% confidence interval. For a more realistic representation of the MARs, the results were

smoothed using the loess smoothing technique, with tricube weighting and polynomial regression using the SigmaPlot software version 11.0 (sampling proportion = 0.100; polynomial degree = 1). The grey lines represent the raw, unsmoothed MARs while the black lines show the smoothed MAR values.

Fig. S6. Dust mass accumulation rate (MAR) as a function of age for the LPSs in Croatia. The MARs were attained using the r.bacon model. The curves show the MARs according to the mean ages within 95% confidence interval. For a more realistic representa-

tion of the MARs, the results were smoothed using the loess smoothing technique, with tricube weighting and polynomial regression using the SigmaPlot software version 11.0 (sampling proportion = 0.100; polynomial degree = 1). The grey lines represent the raw, unsmoothed MARs while the black lines show the smoothed MAR values.

Data S1. Data set presenting r.bacon age-depth model data, calculated sedimentation rates (SRs), mass accumulation rates (MARs), smoothed SigmaPlot data, and ages used for r.bacon modelling.